A Real Cost of Free Trades: Retail Option Trading Increases the Volatility of Underlying Securities*

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This draft: February 11, 2026

Abstract

We study how retail option trading affects underlying equity markets. Exploiting Robinhood's introduction of commission-free option trading, we show that retail trading in options raises the volatility of underlying stocks. We trace this effect to option market makers' dynamic hedging of retail order flow: retail trades are more likely to be intermediated by dealers and to involve higher-gamma contracts, for which dealer hedging in the underlying reinforces price movements. Consistent with retail order flow being relatively uninformed, we show that the underlying security's liquidity improves and price discovery decreases. We establish causality using two difference-in-differences designs.

JEL Classification Codes: G10, G12, G13, G14.

Keywords: Retail traders, Options, Robinhood, Volatility.

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*We thank Dmitriy Muravyev, Neil Pearson, Louis Gagnon, Pedro Garcia-Ares, and the participants at the 2023 Derivatives and Asset Pricing Conference, the Virtual Derivatives Workshop, the Northern Finance Association meetings, the Financial Management Association meetings, and seminar participants at the University of Cambridge Judge Business School. We gratefully acknowledge funding support from the Richard A. Mayo Center for Asset Management.

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1 Introduction

Volatility is a central concern in asset pricing, risk management, and market design. It influences trading costs, margin requirements, and the speed with which prices incorporate information. Yet shifts in volatility may originate not only from fundamentals, but also from trading activity. Understanding how the trading of specific investor groups contributes to volatility is therefore essential to evaluating the effects of market innovations on market quality, especially as broader swaths of the population actively trade securities and regulators increasingly debate the merits of expanding retail investor access to sophisticated instruments. This paper examines one such innovation—the introduction of low cost option trading to retail traders—and shows that it increases the volatility of underlying equity prices.

Retail investors are attractive market participants: they are plausibly uninformed about fundamental values and the trading costs they incur. Unsurprisingly, market design increasingly caters to retail activity, whether upfront through the rise of zero-fee brokerages and off-exchange execution (Dyhrberg, Shkilko, and Werner, 2025), or within the trading process through payment for order flow (Ernst and Spatt, 2022; Bryzgalova, Pavlova, and Sikorskaya, 2023).

The recent expansion of retail participation in options markets has prompted growing concern about retail trader welfare. Prior research documents that retail option traders trade poorly (Bryzgalova et al., 2023), face high spreads (Bogousslavsky and Muravyev, 2024), and frequently exercise sub-optimally (Bryzgalova, Pavlova, and Sikorskaya, 2025).[1] As to market quality—and thus to the effect on market participants more broadly—the impact of retail trading is not yet fully understood. While uninformed retail trading may improve market quality by lowering (relative) adverse selection costs, herding and other behavioral biases may disrupt prices.

In this paper, we document the impact of retail option trading on the underlying equity market. We make three contributions. First, we show that retail option trading increases the volatility of the underlying securities, causally linking investor order flow in option markets to the behavior of stock market prices. Second, we provide evidence

[1] See also Bauer, Cosemans, and Eichholtz (2009); Ernst and Spatt (2022); de Silva, Smith, and So (2022); Beckmeyer, Branger, and Gayda (2023); Bogousslavsky and Muravyev (2024); Wang, Jones, and Zhang (2024); Agarwal, Ghosh, Prabhala, and Zhao (2025); Avila (2025).

that option market making creates the link between the two markets. Third, we show that increased retail *option* trading improves *stock* market liquidity but decreases its price informativeness. This impact is consistent with models that assume retail order flow is uninformed.

Our empirical strategy focuses on the sharp fee reduction experienced by retail traders when Robinhood added option trading to its fee-free trading platform in December 2017. Retail traders who previously paid fees ranging from $6 to $11 per transaction could now trade for free. As expected, we show that the volume of single-contract options (a proxy for retail option trading) increased by over a third. More importantly, we show that the increase in small trades coincided with an increase in volatility—though, of course, this contemporaneous evidence is only suggestive.

To establish causality, we employ two differences-in-differences designs that compare securities identical in fundamentals but differently exposed to the Robinhood shock. First, we compare US stocks to their Canadian cross-listings, which were unaffected by the shock as Robinhood only allows the trading by US residents of U.S.-dollar-denominated stocks and their options. Second, we compare dual-class shares of the same firm, where only one share class underlies listed option contracts. Across both settings, we find robust evidence that the reduced retail trading cost for option contracts increase the volatility of the underlying security.

We complement our causal tests with cross-sectional evidence from a comprehensive set of U.S. stocks. For this sample, we show that similar results obtain when we condition on ex ante characteristics that predict whether a company's options are more likely to be affected by the fee reduction. These characteristics capture the relative cost of obtaining exposure through call options versus purchasing the underlying stock. For example, when call options are inexpensive relative to the share price, a fee reduction is more likely to shift retail demand toward options (an approach supported by the evidence by Agarwal et al., 2025). Across four such measures of a stock's propensity to be affected, we find the expected pattern: stocks more exposed to the fee reduction experience larger increases in retail option trading and volatility in the period following the Robinhood event. The magnitudes are economically significant. In the lowest quintile—where options are relatively cheap—volatility increases by roughly 35%. In the highest quintile, the increase is approximately 3%.

Because we argue that retail option trading alters the nature of volatility in the underlying securities, these effects should not appear only in very short-horizon measures. Indeed, we find increases in daily volatility based on 5-, 10-, 30-, and 60-minute returns, as well as weekly and monthly volatility based on daily returns.

The mechanism linking retail option activity to higher underlying volatility is a specific instance of the dynamic-hedging channel emphasized by Ni, Pearson, Poteshman, and White (2021), building on Golez and Jackwerth (2012) and Ni, Pearson, and Poteshman (2005). When option market makers hedge their positions, their trades amplify volatility whenever their portfolio gamma is negative, a prediction consistent with theoretical models (Frey, 2000; Wilmott and Schönbucher, 2000). We provide evidence consistent with this mechanism: retail trading lowers market makers' gamma through imbalances that disproportionately demand high gamma contracts.

Having established a link between retail option trading and underlying-stock volatility, we examine broader implications for market quality. While we emphasize a non-informational channel, as dynamic hedging drives the excess movement in prices, the associated order flow is largely uninformed, which should improve liquidity (Kyle, 1985). Consistent with this prediction, we document declines in quoted spreads, effective spreads, realized spreads, and price impact around the fee reduction.

Reinforcing the notion that retail order flow is uninformative, we find a decline in the share of price discovery occurring in the affected U.S. markets, using the methodology developed by Hasbrouck (1995). After Robinhood introduces fee-free option trading, Canadian cross-listings contribute relatively more to price discovery than their U.S. counterparts. We find analogous results in the dual-class share setting.

Taken together, our results suggest an increase in retail participation increases the volatility of underlying securities and reduces their price informativeness relative to unaffected markets. There is vigorous debate in the United States regarding altering retail investors' access to options, while regulators in other countries have acted aggressively to curb retail option trading.[2] Our results speak to the potential impact

[2]For the SEC, see https://www.sec.gov/newsroom/speeches-statements/crenshaw-2021-09-27 and https://www.sec.gov/rules-regulations/2021/08/request-information-comments-broker-dealer-investment-adviser-digital-engagement-practices-related. For details on an Indian regulation, see https://www.bloomberg.com/news/articles/2025-09-30/india-to-launch-type-of-options-that-now-dominates-us-trading.

of regulations aimed at affecting household access to sophisticated financial assets, highlighting consequences beyond the direct effect on trader welfare.

The remainder of the paper is organized as follows. In Section 2, we describe the data and variables we use in our tests. Section 3 presents our main results, focusing on ex ante measures, and two difference-in-difference settings. In Section 4 we focus on the channels at play. We detail the impact on market liquidity and price informativeness of the increased idiosyncratic volatility in Section 5. Section 6 concludes.

Literature Review

Our study is related to several areas of research. The first explores how retail trading affects stock-market quality. Retail investors can provide liquidity (Kaniel, Saar, and Titman, 2008; Barrot, Kaniel, and Sraer, 2016; Glossner, Matos, Ramelli, and Wagner, 2025; Ozik, Sadka, and Shen, 2021; Eaton, Green, Roseman, and Wu, 2022), but may also act as noise traders and increase volatility (Brandt, Brav, Graham, and Kumar, 2010; Foucault, Sraer, and Thesmar, 2011). We contribute by showing that retail investors amplify stock volatility *indirectly* through the option market. Because options embed leverage, retail flow in options can magnify the volatility impact of retail activity relative to trading in the underlying.

A second strand focuses on retail option trading.[3] Recent papers highlight the poor performance of retail option traders (Bryzgalova et al., 2023; Hu, Kirilova, Park, and Ryu, 2024), the impact of retail option order flow on option prices (Eaton et al., 2022), and the rents earned by market makers and payment-for-order-flow intermediaries (Bryzgalova et al., 2023; Ernst and Spatt, 2022). de Silva et al. (2022) show that retail option trades concentrate around earnings announcements, often to the investors' detriment. While these studies focus on the costs and consequences of option trading for retail traders and market makers, we examine how retail option trades spill over to the underlying equity. Our results complement Eaton et al. (2022), who document an effect on implied volatility; because single-stock variance swaps are unavailable,

[3] Earlier work studying the drivers and profitability of retail option trading includes Lakonishok, Lee, Pearson, and Poteshman (2007); Lemmon and Ni (2008); Bauer et al. (2009); Choy and Wei (2012); Choy (2015); Dorn, Dorn, and Sengmueller (2015).

the difference between implied and realized volatility cannot be arbitraged and, thus, the two quantities can develop independently.

A third strand examines how option trading influences underlying securities. Early work documents price patterns around expirations driven by hedging flows (Klemkosky, 1978; Ni et al., 2021), and a large literature shows that option order flow informs underlying prices (Ni et al., 2005; Roll, Schwartz, and Subrahmanyam, 2010; Hu, 2014; Ge, Lin, and Pearson, 2016; Chordia, Kurov, Muravyev, and Subrahmanyam, 2021; Weinbaum, Fodor, Muravyev, and Cremers, 2023). Evidence on volatility effects is mixed: Conrad (1989) find declines following option introductions, while Bollen (1998) attribute these results to timing. More recent work links market-maker rebalancing to higher (Ni et al., 2021) and lower (Adams, Dim, Eraker, Fontaine, Ornthanalai, and Vilkov, 2025) volatility. We extend this literature by providing causal evidence—using optioned/optionless share classes and U.S.–Canada cross-listings—that retail-driven option trading raises underlying-stock volatility and by tying this effect to dealers' hedging of retail gamma exposure.

Finally, our work connects to a growing literature studying the welfare implications of giving retail investors access to leverage (Heimer and Simsek, 2019; Heimer and Imas, 2022) and complex financial instruments (Knüpfer, Rantala, and Vokata, 2021; Vokata, 2021). We contribute by showing that reducing frictions to retail derivatives trading not only affects retail investors' own outcomes, but also generates negative externalities for the broader equity market through its impact on volatility and price discovery.

2 Data

This section details the measures we employ and the data sourced to calculate them. Our main sample spans June 1, 2017, to May 31, 2018, covering the six months before and after Robinhood's introduction of option trading on December 13, 2017. The unit of observation in most specifications is the stock–day. We study both stock-market outcomes—volatility, trading volume, and liquidity—and option-market activity—including trading volume, retail trading intensity, and option characteristics.

Our baseline universe consists of all U.S.-listed common stocks and exchange-traded funds (ETFs) that serve as underlying securities for traded options. In some analyses, we include U.S.-incorporated firms without listed options; in others, we use Canadian firms cross-listed in the United States as a control group. Section 2.1 describes the option data we use, and Section 2.2 details the construction of stock-market measures.

2.1 Option Retail Trading

We obtain detailed transaction-level option data from the Chicago Board Options Exchange (CBOE). The data cover all option trades executed on U.S. option exchanges, provided the contract is listed on CBOE. We therefore observe all trades for options written on individual stocks and ETFs (but not index options) across the 16 U.S. option exchanges. For each trade, we observe price, size, executing exchange, NBBO quotes, and the contemporaneous price of the underlying security.

We construct a measure of retail option trading by focusing on 1-contract transactions, the smallest available to traders, as fractional trades are not available for options.[4] We complement transaction data with contract-level end-of-day summaries from CBOE, which report implied volatility and option greeks. We focus on Delta ($\Delta = \frac{\partial o}{\partial S}$) and Gamma ($\Gamma = \frac{\partial \Delta}{\partial S} = \frac{\partial^2 o}{\partial S^2}$), where S is the underlying's price, and o and Δ the option's price and delta, respectively. To characterize the relative attractiveness of options versus stocks, we compute option omega, a measure of embedded leverage, $\Omega = \left| \frac{\partial o}{o} \left(\frac{\partial S}{S} \right)^{-1} \right| = \frac{|\Delta| S}{o}$, following Frazzini and Pedersen (2022).

We further use the Nasdaq Options Trade Outline (NOTO) and PHLX Options Trade Outline (PHOTO), which provide trader-type and trade-size information for transactions executed on the Nasdaq Options Market and Nasdaq PHLX. These data report signed volume and changes in open interest for five trader categories—firm proprietary traders, brokers and dealers, market makers, regular customers, and professional

[4] Bryzgalova et al. (2023) develop a novel measure of retail option trading based on a price-improvement flag (SLAN, "single-leg price improvement mechanism"). The SLAN indicator is available only from November 2019 onward, which postdates Robinhood's introduction of option trading and therefore falls outside our main sample. In Appendix Section A.1, we show that, for the 2019–2020 period in which our data overlap with data made available by Bryzgalova, Pavlova, and Sikorskaya, our proxy closely aligns with the measure, yielding a very similar ranking of stocks by retail option trading intensity.

customers—and three trade-size bins (up to 100 contracts, 100–199 contracts, and 200 or more). For each trader, contract, and day, the data record buy-to-open, buy-to-close, sell-to-open, and sell-to-close activity. We provide further details on how we employ these quantities to link dealers' gamma to retail trading in Section 4.

Figure 1 plots the evolution of aggregate, market-wide trading volume attributable to one-contract option trades, both in levels and as a share of total option volume over four years. Retail option trading grows most aggressively in 2020, yet the series exhibits a clear level and trend shift around Robinhood's introduction of free option trading on December 13, 2017. Figure 2 focuses on our study period and shows a distinct regime shift around the Robinhood shock. Although the transition unfolds gradually over approximately one month, the change is economically large: Retail trading increases by 43% between the first and second half of the sample.

Importantly, this increase in small trades does not appear to reflect a pre-existing trend. Figure 3 compares the evolution of retail option trading with the popularity of option trading at major retail brokerages, measured using Google search activity. Following Da, Engelberg, and Gao (2011) and Da, Engelberg, and Gao (2015), we obtain weekly Google Search Volume Index (SVI) data for the terms "Options Robinhood", "Options TD Ameritrade", "Options Schwab", and the broader "Option Trading". The SVI series are normalized so that the maximum search intensity across terms and weeks in the sample equals 100, allowing for cross-sectional and time-series comparisons.[5]

The figure provides two validations for our measure and its relation to Robinhood's introduction of free option trading. First, Google searches for "Options Robinhood" display a clear break coinciding with Robinhood's introduction of option trading, while searches related to option trading on other brokerages remain flat until the broad retail trading surge in 2020. Second, prior to 2020, the general search "Option Trading" peaks precisely in the week Robinhood introduces options, and is as popular as "Robinhood Trading", despite being brokerage-agnostic. Taken together, these patterns support the interpretation that one-contract option volume captures retail

[5] The Google Trends data used in this figure are available at `https://trends.google.com/trends/explore?date=2016-01-01%202020-09-30&geo=US&q=options%20robinhood,options%20td%20ameritrade,options%20schwab,option%20trading&hl=en`.

trading activity and that the observed increase is directly linked to the availability of option trading on Robinhood.

2.2 Other data

We use Trade and Quote (TAQ) data to construct stock-market measures, following Holden and Jacobsen (2014). From TAQ, we compute trading volume, order imbalance, liquidity, and volatility. The relative quoted spread at time s is

$$QuotedSp_s = 100\frac{A_s - B_s}{M_s},$$

where A_s and B_s denote the national best ask and bid, and M_s the midpoint. The relative effective spread, realized spread, and price impact for trade k are defined as

$$EffectiveSp_k = 100\frac{2D_k(P_k - M_s)}{M_s}, \quad RealizedSp_k = 100\frac{2D_k(P_k - M_{s+5})}{M_s}$$

$$PriceImpact_k = 100\frac{2D_k(M_{s+5} - M_s)}{M_s},$$

where D_k equals 1 (–1) for buyer- (seller-) initiated trades (Lee and Ready, 1991), P_k is the trade price, and M_s and M_{s+5} are the midpoints at the time of the trade and five minutes later. We aggregate these measures to the stock-day level using dollar-volume weighting for trade-based measures and time weighting for quote-based measures.

We compute daily volatility using excess returns over the market, proxied by the SPY ETF, at 5-, 10-, 30-, and 60-minute intervals. Retail stock trading volume is measured following Boehmer, Jones, Zhang, and Zhang (2021). For each FINRA-reported trade (exchange code "D" in TAQ), we compute the fractional penny component $Z = 100 \bmod (P, 0.01)$ and classify trades as retail buys (sells) if $0.6 < Z < 1$ ($0 < Z < 0.4$). We aggregate retail volume to the stock-day level.

We obtain stock characteristics from the Center for Research in Security Prices (CRSP). Trades and quotes for Canadian firms cross-listed in the U.S. come from the Toronto Stock Exchange (TSX) and include nanosecond-timestamped best-bid and ask updates. We obtain intraday USD–CAD exchange rates at the five-minute frequency from Olsen Financial Technologies.

3 Increase in Retail Option Trading and Volatility

In this section, we show that increased retail participation in the option market leads to higher idiosyncratic volatility of the underlying securities. Section 3.1 presents preliminary evidence based on ex-post sorting over changes in retail option trading. In Section 3.2, we obtain our main result by sorting stocks by their likelihood to be treated, i.e., by sorting them by characteristics that ex-ante predict whether a decrease in option trading cost will elicit higher option retail trading in a specific ticker. Sections 3.3 and 3.4 provide causal evidence using two difference-in-differences designs. In the first, we compare U.S.– and Canadian-dollar–denominated shares of cross-listed firms, where only the former are affected by U.S. option trading. In the second, we exploit firms with dual share classes, only some of which underlie listed option contracts.

3.1 Preliminary Evidence

We begin by documenting a positive relation between retail option trading in a ticker and the stock's idiosyncratic volatility. Using the full sample of optioned securities, we estimate

$$Volatility_{it} = \alpha_i + \alpha_t + \beta\, RetailOVolume_{it} + \varepsilon_{it}, \tag{1}$$

where $Volatility_{it}$ is the logarithm of the standard deviation of five-minute stock returns net of market returns for stock i on day t. $RetailOVolume_{it}$ denotes the logarithm of day-t trading volume accounted for by one-contract option trades, aggregated across all options that have stock-i as underlying. The specification includes stock and day fixed effects, and standard errors are clustered by stock and day.

Table 1 reports the results. Retail option trading is positively and statistically significantly related to idiosyncratic volatility. A 10% increase in retail option volume is associated with a 1% increase in volatility; alternatively, a one-standard-deviation increase in retail option volume raises volatility by approximately 0.2 standard deviations. This result is not driven by overall option market activity. Controlling for total option volume in Specification 2 leaves the coefficient on retail option trading largely unchanged, and the effect of retail volume is economically larger than that of total option volume. Normalizing the former by the latter, as in Specification 3, does not alter the conclusion.

Because retail trading in the stock market has been shown to increase volatility (Foucault et al., 2011), Specification 4 controls for both retail ($RetailSVolume_{it}$) and total ($SVolume_{it}$) stock trading volume. The coefficient of interest remains stable. In Specification 5, we include industry-by-day fixed effects to account for correlated retail trading across related firms in the same industry (Welch, 2022). The estimates are again virtually unchanged.

While these results establish a strong correlation between retail option trading and volatility, they clearly do not show a causal relation. To achieve identification, we exploit Robinhood's introduction of option trading, which sharply reduced retail option trading costs—from \$6–\$11 per transaction to zero. In the next subsection, we show that ex ante measures capturing a stock's propensity to be affected by this cost reduction predict subsequent increases in both retail option trading and volatility—that is, stocks for which option trading volume should increase the most following a cut in trading costs exhibit both higher retail option trading and volatility. We then provide causal evidence using difference-in-differences designs based on cross-listed and dual-class shares.

Before doing so, we first demonstrate that there exists an ex-post change in markets consistent with our maintained hypotheses. We classify stocks based on the increase in one-contract option trades in the three months before and after Robinhood's option introduction, excluding the month immediately prior the event. Panel A of Figure 4 plots scaled idiosyncratic volatility separately for stocks in the top tercile of retail option trading growth and for those in the remaining terciles. Prior to December 2017, volatility evolves similarly across the two groups. After the introduction of option trading, however, stocks that experienced larger increases in retail option activity exhibit a more pronounced increase in idiosyncratic volatility.

Table 2 confirms this pattern in a regression framework:

$$Volatility_{it} = \alpha_i + \alpha_t + \beta\, IncreasedORT_i \times Post_t + \varepsilon_{it}, \tag{2}$$

where $IncreasedORT_i$ is an indicator for stocks in the top tercile of retail option trading growth before and after Robinhood's introduction of option, and $Post_t$ equals one after the event. Specification 4 shows that stocks for which option retail trading increased the most experience a 5% larger increase in volatility relative to other stocks. Consistent with this classification, retail option trading for these stocks rises by 46% more than for the other stocks (Specification 8).

Panel B of Figure 4 plots the coefficients from a dynamic specification, $Volatility_{it} = \alpha_i + \alpha_t + \beta_t \, IncreasedORT_i + \varepsilon_{it}$, which traces the evolution of differential volatility over time. The estimates confirm the absence of pre-trends and the emergence of a persistent volatility increase following the introduction of option trading.

3.2 Sorting on Ex-ante Stock Characteristics

We hypothesize that, *ceteris paribus*, retail investors respond more strongly to a reduction in option trading costs when the cost of obtaining exposure through options is low relative to the underlying stock price. In such cases, a decline in commissions represents a more salient change and should induce a larger increase in retail option trading—and, in turn, a larger increase in underlying-stock volatility.

To test this hypothesis, we sort stocks by their average implied volatility (by their underlying stock price) in the three months preceding Robinhood's introduction of option trading. Stocks with lower implied volatility (higher stock prices)—those with relatively cheaper option contracts—should experience larger increases in both retail option trading and volatility following a fee reduction.

Measures based solely on implied volatility or stock price, however, do not capture retail investors' contract selection. Retail traders can lower option costs by choosing short-maturity or out-of-the-money contracts for a stock with a given price and implied volatility. We therefore consider the ratio of the average price of one-contract (retail) option trades to the underlying stock price. Stocks for which retail investors select relatively cheaper option contracts should be more strongly affected by a fee reduction. Finally, because investors may seek out the leverage embedded in options, we also sort stocks by the average embedded leverage (omega) of their options. We expect stocks for which retail investors select higher-omega contracts to experience larger increases in retail option activity and volatility after a fee reduction.

Table 3 reports descriptive statistics that support these predictions. For each variable, we compute the stock-level median over the pre-event period and report cross-sectional averages separately for stocks with large and small increases in retail option trading. Stocks with larger increases tend to have lower implied volatility, higher stock prices, and lower average one-contract option prices.

To formally test these hypothesis, we estimate:

$$Volatility_{it} = \alpha_i + \alpha_t + \beta\, LowIV_i \times Post_t + \varepsilon_{it}, \tag{3}$$

where $LowIV_i$ (and analogously $HighS_i$, $LowO/S_i$, and $HighOmega_i$) equals one if stock i falls in the bottom (or top) tercile of the corresponding characteristic, measured over the three months prior to Robinhood's option introduction, and zero otherwise.

Table 4 reports the results. Consistent with our hypothesis, stocks for which the reduction in option commissions is most salient experience the largest increases in both idiosyncratic volatility and retail option trading. For example, stocks in the lowest tercile of implied volatility exhibit a roughly 11% larger increase in volatility and a 10% larger increase in retail option volume relative to other stocks over the twelve months surrounding the event. Similarly, stocks for which retail investors select the highest embedded leverage experience increases of approximately 9% in volatility and 8% in retail option trading around the event.

These findings are not sensitive to the choice of quantiles or functional form. Table 5 repeats the analysis using quintile of the sorting variables and show the average changes in volatility and retail option trading around Robinhood's introduction of options. Panel A reports one-way sorts based on the option-to-stock price ratio and embedded leverage, while Panels B and C present two-way sorts based on implied volatility and stock price. Stocks in the lowest quintile of the retail option price–to–stock price ratio experience an increase in volatility of roughly 35%, compared with an increase of only about 3% for stocks in the highest quintile.[6]

To assess the statistical robustness of the estimates in Eq. (3), we adopt a nonparametric bootstrap approach. We re-sort stocks by the same characteristics for each

[6] Appendix Table A-1 replicates these results in a regression framework and confirms that the differences are not only economically but also statistically significant.

pre-pseudo-event window in the sample and re-estimate Eq. (3) for each subsample.[7] Appendix Figure A-3 and Table A-2 report the resulting coefficient distributions and confirm the robustness of our inferences—all estimates are highly statistically significant using these alternative confidence estimates. The bootstrapping exercise additionally provide evidence that sorting stocks over the ex-ante characteristics we selected does not regularly predict future increases in volatility.

Finally, the effects we document are not high-frequency microstructure noise, as our results are robust to measuring volatility at much lower frequencies. Table 6 replicates the analysis using volatility measured at 10-, 30-, and 60-minute intervals, as well as weekly and monthly volatility based on open-to-close returns. The magnitude of the volatility increase remains remarkably stable across time horizons, consistent with a structural shift in volatility rather than a transitory effect.

3.3 Cross-listed Stocks

To identify the causal effect of retail option trading on volatility, we exploit firms that list economically identical shares in both the United States and Canada. Our sample consists of 71 companies whose U.S.- and Canadian-listed shares represent the same claim on cash flows but trade in different currencies. Unlike American Depositary Receipts, these shares are fully fungible aside from currency denomination. Importantly, Robinhood allows trading in the U.S.-dollar–denominated shares and in options written on them, but does not offer trading in Canadian-dollar–denominated shares or in options listed on Canadian exchanges. For example, Robinhood allows trading in Precision Drilling Corporation's U.S.-listed shares (NYSE:PDS) and the corresponding CBOE options, but not in its Canadian-listed shares (TSX:PD) or Montreal Exchange options.

This institutional feature allows us to compare the evolution of volatility for U.S.- and Canadian-denominated shares of the same firm around Robinhood's introduction of

[7] To avoid sampling days too far away from the actual event, while still generating enough draws, we limit the event-study window to 90 (rather than six months) days before and after each pseudo-event date. We calculate new rankings based on the first half of each sample. We generate a new baseline, which we report in Panel A of Table A-2. We do not include bootstrapped samples that straddle the actual event date.

option trading. Because the underlying cash-flow claims are identical, any divergence in volatility around the event—aside from exchange-rate effects—can be attributed to increased retail option trading affecting only the U.S.-listed shares.

Panel A of Figure 5 plots the relative difference in idiosyncratic volatility between U.S.- and Canadian-listed shares. Prior to December 2017, the volatility differential is centered around zero. Following the introduction of option trading on Robinhood, the differential rises to as much as 10%, indicating an increase in excess volatility in the U.S. market. The same panel also reports aggregate U.S. one-contract option volume. The correlation between retail option trading intensity and excess U.S. volatility is 43%, strongly supporting the hypothesis that an increase in the former led to the latter.

We formalize this comparison in a difference-in-differences framework:

$$Volatility_{it} = \alpha_i + \alpha_t + \beta\, US_i \times Post_t + \varepsilon_{it}, \tag{4}$$

where $Volatility_{it}$ is the logarithm of five-minute return volatility for share i on day t, US_i equals one for U.S.-listed shares and zero for Canadian-listed shares, and $Post_t$ equals one after December 13, 2017. Table 7 reports the results. Specification 1 shows that, following Robinhood's option introduction, U.S.-listed shares become approximately 3% more volatile than their Canadian counterparts.

To ensure that exchange-rate fluctuations do not drive this result, we obtain intraday CAD–USD data and compute USD-equivalent prices for Canadian-listed shares. Specification 2 repeats the analysis using volatility measured in USD-equivalent terms. The estimated effect is nearly unchanged, indicating that foreign-exchange volatility explains little of the observed differential. Panels A and B of Figure 6 report dynamic estimates corresponding to Specifications 1 and 2, respectively. In both cases, the volatility differential is negligible prior to the event and increases sharply thereafter.

As an alternative approach, we directly regress the relative volatility difference between U.S.- and Canadian-listed shares on the post-event indicator and daily exchange-rate volatility:

$$Volatility_{it}^{US} - Volatility_{it}^{CA} = \alpha_i + \beta\, Post_t + \gamma\, FXVol_t + \varepsilon_{it}, \tag{5}$$

where $FXVol_t$ denotes intraday CAD–USD volatility. Specification 3 of Table 7 shows that controlling for exchange-rate volatility in this alternative manner leaves the estimate of β largely unchanged. Specification 4 reports the difference in volatility after the CAD-denominated prices have been calculated in USD-equivalents, to unchanged results.

Higher volatility in U.S.-listed shares implies greater noise in the price-formation process, which should increase the absolute price difference between cross-listed shares. We therefore measure deviations as

$$Deviation_{it} = \left| P_{it}^{US} - \frac{P_{it}^{CA}}{FX_t} \right|,$$

and examine both its level and intraday variability. Panel B of Table 7 shows in a regression setting that both the absolute price discrepancy and its dispersion increase after the introduction of retail option trading, and that the result is not driven by exchange-rate considerations. Panel C of Figure 6 shows graphically that these alternative measures of excess volatility also co-move with retail option activity. In Section 5, we show that this increased noise shifts price discovery away from U.S. markets toward Canadian markets.

Finally, we examine heterogeneity in treatment intensity within the interlisted sample using a triple-differences specification. We interact the post-event indicator with US_i and with indicators capturing ex ante company-c characteristics—implied volatility, stock price, option-to-stock price ratio, and embedded leverage—measured prior to the event, similar to the analyses in the previous section. The specification saturates share, time, and interaction fixed effects:

$$Volatility_{it} = \alpha_i + \alpha_t + \alpha_c \times Post_t + \alpha_t \times US_i + \beta\, US_i \times Post_t \times LowIV_c + \varepsilon_{it}. \quad (6)$$

Table 8 reports the results. In this analysis, we are comparing the relative increase in relative volatility between USD- and CAD-denominated shares between two cross-listed companies. Consistent with our earlier findings, U.S.-listed shares of firms with ex ante cheaper options (lower implied volatility) experience a 4% larger increase in volatility relative to their Canadian counterparts, as compared to firms with more expensive options.

3.4 Dual-class Shares

We employ a complementary identification strategy using U.S. firms with multiple share classes. Twenty-two firms list dual share classes on U.S. exchanges, but only one class underlies listed option contracts.[8] Because the two share classes represent claims on the same firm and trade simultaneously, this setting provides a second measure of excess volatility: the difference in volatility between optioned and optionless share classes.

Panel B of Figure 5 plots the average volatility differential between optioned and optionless shares, along with aggregate one-contract option volume. As in the cross-listed sample, excess volatility closely tracks the intensity of retail option trading.

We estimate the corresponding difference-in-differences model:

$$Volatility_{it} = \alpha_i + \alpha_t + \beta\, Optioned_i \times Post_t + \varepsilon_{it}, \tag{7}$$

where $Optioned_i$ equals one for the share class underlying listed options. Specification 1 of Table 9 shows that, following Robinhood's introduction of option trading, optioned share classes become approximately 3% more volatile than their optionless counterparts. Panel C of Figure 6 report dynamic estimates corresponding to Specifications 1: the volatility differential is negligible prior to the event and increases thereafter.

4 Market Maker's Hedging Demand and Retail Trading

The previous sections establish that the expansion of retail option trading increases the volatility of the underlying stocks. In this section, we present evidence consistent with this increase in volatility being driven by market makers' hedging of their exposure to retail option trading.

[8] Liberty Global lists three share classes, two of which underlie option contracts.

We first examine market makers' option positions in Section 4.1. We show that market makers' net gamma becomes more negative as retail option trading expands, consistent with dynamic hedging being the mechanism at play: lower gamma implies that market makers' hedging trades amplify price movements in the underlying (Ni et al., 2021). We then show that market maker trades offset retail trading imbalances, directly linking market maker positions to retail order flow.

In Section 4.2, we document several characteristics of retail option trading that help explain why expanded retail activity reduces market makers' gamma. First, retail traders herd: their buy and sell orders are less likely to offset one another than those of other market participants, generating imbalances that must be absorbed by market makers. Second, retail option trading is disproportionately tilted toward high-gamma contracts. Third, retail trading behavior itself changes around the Robinhood event, further strengthening retail investors' preference for high-gamma exposure.

Consistent with the existing literature on option-induced volatility, we focus primarily on market makers' gamma exposure, which determines whether dynamic hedging trades amplify or dampen volatility. In Section 4.3, we consider an alternative channel—initial (delta) hedging—and show that it is unlikely to be the primary driver of the volatility increase.

4.1 Market Maker's Gamma

Option market makers typically engage in dynamic delta hedging, trading the underlying asset as the delta of their option portfolio evolves due to changes in the underlying price, the passage of time, or shifts in implied volatility. When the underlying market is not perfectly liquid, these hedging trades affect prices. Ni et al. (2021) show empirically that underlying-stock volatility is decreasing in option market makers' portfolio gamma.[9]

When investors predominantly buy options, dealers' portfolios exhibit negative gamma. In this case, market makers buy the underlying following price increases and sell following price declines, amplifying volatility. When dealers hold positive gamma, their

[9] This prediction had previously been formalized in theoretical work by Jarrow (1994); Frey and Stremme (1997); Frey (1998); Platen and Schweizer (1998); Ronnie Sircar and Papanicolaou (1998); Frey (2000); Wilmott and Schönbucher (2000).

hedging trades instead dampen price movements. In this section, we examine whether increased retail option trading pushes market makers' gamma toward more negative values.

We measure market makers' hedging exposure using end-of-day position data from the PHLX Options Trade Outline (PHOTO) and Nasdaq Options Trade Outline (NOTO). These data provide trader-type and trade-size information, recording for each trader category—firm proprietary traders, brokers and dealers, regular customers, and professional customers—the number and volume of buy-to-open, buy-to-close, sell-to-open, and sell-to-close transactions. Trades are aggregated by contract, trader type, and trade-size category (up to 100 contracts, 100–199 contracts, and 200 or more). For market makers, the data similarly report aggregate buy and sell activity.

Following Ni et al. (2021), we infer market makers' net open interest for option series j at the end of day t, $NetOpenInterest_{j,t}$, by aggregating the positions of all non–market-maker investors:

$$
\begin{aligned}
OpenInterest_{j,t}^{Buy,y} &= OpenInterest_{j,t-1}^{Buy,y} + Volume_{j,t}^{OpenBuy,y} - Volume_{j,t}^{CloseSell,y}, \\
OpenInterest_{j,t}^{Sell,y} &= OpenInterest_{j,t-1}^{Sell,y} + Volume_{j,t}^{OpenSell,y} - Volume_{j,t}^{CloseBuy,y}, \\
NetOpenInterest_{j,t} &= -\left[\sum_{y \in Public}\left(OpenInterest_{j,t}^{Buy,y} - OpenInterest_{j,t}^{Sell,y}\right)\right],
\end{aligned}
$$

where open interest is initialized at zero on the first trading day of each option series. $Volume_{j,t}^{OpenBuy,y}$ and $Volume_{j,t}^{OpenSell,y}$ denote trades establishing new long or short positions by investor type y, while $Volume_{j,t}^{CloseBuy,y}$ and $Volume_{j,t}^{CloseSell,y}$ denote trades closing existing positions. The set $Public$ includes all investor types other than market makers.

We aggregate across option series for a given underlying asset to construct market makers' net gamma exposure:

$$
net_t\Gamma = 100\frac{S_t}{M_t}\sum_j NetOpenInterest_{j,t}\Gamma_j,
$$

where S_t is the underlying stock price and M_t is the number of shares outstanding.

This scaling follows the literature and facilitates comparisons across stocks. We obtain Γ_j from OptionMetrics. We also compute Black–Scholes gamma independently and find it to be nearly identical, and therefore rely on the publicly available measure.

We test whether retail option trading affects market maker gamma around the Robinhood event using the ex ante sorts from earlier sections. If retail order flow drives down market maker gamma, the effect should be strongest for stocks that are more attractive to retail traders. We estimate specifications analogous to Eq. (3), replacing the dependent variable with measures of market makers' gamma exposure. Table 10 reports the results.

In Panel A, the dependent variable is an indicator equal to one if market makers' net gamma is negative. Stocks that are ex ante more likely to be affected by the fee reduction exhibit a 2–3% higher probability of market makers holding negative-gamma portfolios. Given that the unconditional probability of negative gamma is approximately 45% in our sample, this effect is economically meaningful. Panel B reports similar results using continuous gamma measures.

Although transaction-level counterparty identities are unavailable, several pieces of evidence suggest that retail trades are intermediated primarily by market makers. First, if customers trade predominantly with dealers rather than with one another, customer and dealer gamma exposures should be negatively related. Indeed, PHOTO and NOTO data show that, at the ticker-day level, customers' and dealers' gamma positions are negatively correlated at approximately −78%.

Second, merging position data with order flow, we expect market makers' buy (sell) orders to be negatively related to customer sell (buy) imbalances. We regress market makers' buy (sell) volume on customer sell (buy) volume at the ticker–day level and report the results in Table 11. Specification 1 shows that dealer buy volume is positively related to seller-initiated customer trades, consistent with dealers absorbing customer order flow. The coefficients on retail trade variables exceed those on total trade variables, indicating that dealers are more likely to intermediate small trades. Specification 2 presents the symmetric result for dealer sells.

In Specification 3, we show that daily changes in market makers' gamma are negatively related to customer trade imbalances, further confirming that dealers are the natural counterparties to customers. Moreover, a one-contract increase in retail volume has a larger effect on dealer gamma than an equivalent increase in non-retail volume. We next examine why retail trading has a disproportionate impact on market makers' gamma.

4.2 What Makes Retail Trading Special

The previous subsection shows that market maker gamma declines with retail option trading, particularly for stocks more exposed to the Robinhood shock. In this section, we document three characteristics of retail option trading that help explain this pattern.

First, retail option trading exhibits stronger herding behavior. We examine correlations between buy and sell order flow by sorting contract–day observations into deciles based on absolute trading imbalance and computing correlations separately for retail and non-retail trades. Table 12 shows that, across all but one decile, the correlation between non-retail buys and sells exceeds that between retail buys and sells. In most deciles, non-retail buy and sell volumes are nearly perfectly correlated, whereas the corresponding correlation for retail trades is closer to 0.75. These results indicate that retail orders are less likely to offset one another and are therefore more likely to generate imbalances absorbed by market makers.

Second, retail investors disproportionately buy high-gamma contracts. Table 13 reports contract–day regressions of retail trade imbalances, defined as $\frac{RetailBuys - RetailSells}{RetailBuys + RetailSells}$, on contract gamma and price. We estimate analogous regressions for non-retail trades and for the difference between retail and non-retail imbalances. Retail investors buy significantly more high-gamma options than non-retail investors, pushing market makers' gamma further into negative territory. All specifications include underlying-by-date fixed effects, so identification comes from variation across contracts within the same underlying and day. While beyond the scope of this paper to determine why retail investors prefer high-gamma options, we should notice that this preference is consistent with investors favoring low-priced contracts: High-gamma options have significantly lower prices, as shown in Appendix Figure A-4.

Third, retail trading behavior itself changes around the Robinhood event. Rather than simply scaling up trading uniformly across contracts, retail investors increasingly concentrate in higher-gamma options. Table 14 sorts contracts into gamma quintiles and examines changes in retail order imbalances around the event. With the exception of the lowest gamma quintile, retail imbalances increase monotonically across quintiles, shifting from significantly negative to significantly positive. This pattern is consistent with either an intensive-margin shift in preferences or an extensive-margin increase concentrated in high-gamma contracts; both mechanisms imply a decline in market makers' gamma as dealers absorb the increased retail demand.

To summarize retail investors' contribution to market makers' hedging demand, we compute aggregate gamma demand by investor type. We calculate signed order imbalances for retail (one-contract) and non-retail trades at the contract level and aggregate them to the underlying–day level after weighting by contract gamma. Table 15 shows that customer gamma demand increases significantly around Robinhood's option introduction, but only for small trades. Specifications 4–6 in Table 13 confirm that the propensity of small trades to concentrate in high-gamma contracts strengthens after the event. Figure 7 illustrates these patterns at the market level: gamma demand from large trades remains flat, while gamma demand from small trades trends upward for both calls and puts.

Even though delta demanded from large trades remains stable around the event, the positive (negative) delta demanded by small trades in calls (puts) increases significantly. We therefore turn to the question of whether initial hedging could be driving the observed volatility effects.

4.3 Initial (Delta) Hedging Evidence

While much of the literature emphasizes gamma hedging as the link between option trading and underlying volatility, initial (delta) hedging provides an alternative channel. An increase in retail option trading may be accompanied by greater variability in retail order flow, potentially inducing repeated initial hedging by market makers.

We first examine whether retail investors systematically trade high-delta options. In Table 16, we regress the fraction of option volume attributable to retail trades,

$Retail\%_{it}$, on option characteristics, including price, strike price, and time to expiration. In Specifications 1 and 2, we control for strike price and include underlying-by-trade date-by-expiration fixed effects. Retail investors trade higher-strike calls and lower-strike puts, consistent with a preference for higher-delta options. In Specifications 3 and 4, controlling for time to expiration and including underlying-by-strike-by-trade date fixed effects, we find that retail investors trade longer-maturity options more frequently. Together, these results indicate a preference for high-delta options along both the moneyness and maturity dimensions.

Despite this preference, the evidence does not support initial delta hedging as a primary driver of the volatility increase. Table 15 shows that market makers' aggregate delta exposure responds around Robinhood's option introduction, but the response is concentrated in exposure originating from small trades. Increases in delta exposure from call and put positions largely offset one another, leaving the net change in aggregate dealer delta statistically indistinguishable from zero.[10]

Finally, although zero-days-to-expiry (0DTE) options have attracted substantial attention in recent discussions of retail trading, retail investors in our sample trade longer-dated options more frequently. As documented by Beckmeyer et al. (2023), 0DTE trading rose sharply only after late 2021 and represents a small share of option activity during our sample period. This evidence underscores the heterogeneity of retail option traders, encompassing both short-term, lottery-seeking traders in more recent years and longer-horizon participants present earlier in the sample.

Taken together, the evidence in this section indicates that market makers' dynamic gamma hedging is the primary channel linking increased retail option trading to higher volatility in the underlying stocks. The most compelling evidence is that stocks identified ex ante as more exposed to retail option trading experience a decline in market makers' gamma following Robinhood's introduction of option trading. Supporting evidence comes from the characteristics of retail trading itself: retail orders are more imbalanced, concentrate in high-gamma contracts, increasingly favor such contracts after the event, and exert a larger effect on market makers' gamma per unit of volume than non-retail trades.

[10]Consistent with this finding, Adams et al. (2025) show that gamma hedging, rather than delta hedging, links retail trading in zero-days-to-expiry options to S&P 500 volatility.

5 Effects of Increased Option Retail Trading on Market Quality

Standard microstructure models deliver clear predictions regarding the effects of increased noise trading on equity-market liquidity and price discovery. If the rise in idiosyncratic volatility we document reflects an increase in uninformed trading—amplified by the leverage embedded in options—information-based models such as Kyle (1985) predict improved liquidity, as noise trading masks informed orders, but reduced price efficiency. In contrast, inventory-based models following Stoll (1978) predict that higher volatility monotonically increases transaction costs.

Using the same difference-in-differences settings as in the previous section, we find evidence consistent with the informational channel: retail option trading, through market makers' hedging, raises uninformed volatility in the underlying market, improving liquidity while impairing price discovery. Subsection 5.1 examines liquidity, and Subsection 5.2 studies price discovery.

5.1 Market Liquidity

To assess the liquidity implications of increased retail option trading, we re-estimate the difference-in-differences specifications using liquidity measures as dependent variables. For the cross-listed sample, we compute liquidity measures for Canadian-listed shares using TSX trades and quotes; for the dual-class sample, we compare TAQ-derived liquidity measures for optioned and optionless share classes within firms. We report the results for the dual-class setting in Specifications 2–5 of Table 9 and for the cross-listed setting in Table 17.

In the dual-class sample, liquidity improves for share classes that underlie option contracts—those directly affected by Robinhood's introduction of option trading—relative to optionless share classes, despite the increase in volatility. For example, quoted bid–ask spreads decline by approximately 6% for optioned share classes. Similar patterns emerge in the cross-listed sample: U.S.-listed shares experience greater liquidity improvements than their Canadian-listed counterparts, with quoted spreads declining by roughly 0.5%. Consistent with an informational interpretation of our findings, increased (levered) noise trading leads to improved market liquidity.

5.2 Price Discovery

In Kyle (1985), an increase in idiosyncratic volatility improves liquidity but reduces market efficiency. While quantifying market efficiency for a single market requires establishing an efficient price, the cross-listed and dual-class settings allow us to test whether the prices of the impacted assets become comparatively less informationally efficient following Robinhood's options introduction.

We quantify the share of price discovery that takes place in each market following Hasbrouck (1995). The paper develops a multi-market vector error-correction model to quantify the proportion of the efficient price innovation attributable to that market (the information share). We calculate information shares at the share-pair–day level, using five-minute level prices.[11]

We regress the information share of the non-treated market on the post-event indicator and report the results in Table 18 for the cross-listed sample and in Table 19 for the dual-class sample. Following Robinhood's introduction of option trading, treated markets contribute 1–3% less to price discovery, while non-treated markets contribute correspondingly more. These findings are consistent with increased noise trading in the affected markets. Specifications 2–5 in Table 18 further show that the decline in price discovery is larger for stocks with a higher ex ante propensity to be affected by the reduction in option trading costs.

5.3 Complementarity between the Stock and Option Market

The reduction in option trading costs alters the relationship between retail trading in stock and option markets. As options become cheaper to trade, retail investors may increasingly view the option market as a substitute for direct stock trading.

[11] Because the Hasbrouck methodology relies on a Cholesky decomposition, information shares are generally bounded rather than point-identified. Following the literature, we use the midpoint of the upper and lower bounds as our estimate.

We quantify this substitution by computing the daily cross-sectional rank correlation between the share of retail trading volume in the option market and the share of retail trading volume in the stock market, using the BJZZ (Boehmer et al., 2021) algorithm to identify retail stock trades. Figure 8 plots this correlation over time. The correlation increases markedly following Robinhood's introduction of option trading, indicating that retail investors increasingly shift activity between the stock and option markets as the cost of trading options declines.

6 Conclusions

This paper studies how retail participation in option markets affects the underlying equity market. While a growing literature examines the welfare consequences of low-cost option trading for retail investors themselves, we document a distinct and economically meaningful spillover: increased retail option trading raises the idiosyncratic volatility of the underlying stocks.

Exploiting Robinhood's introduction of fee-free option trading, we show that retail option activity increased sharply following the platform's action and that this increase was accompanied by a persistent rise in underlying-stock volatility. The effect is strongest for stocks for which the reduction in option trading costs is most salient, both ex post and ex ante. Using two complementary difference-in-differences designs—cross-listed U.S.–Canadian shares and dual-class shares with and without listed options—we provide evidence that the relation is causal.

We trace this effect to the hedging behavior of option market makers. Retail investors disproportionately buy options, trade directionally, and concentrate in high-gamma contracts. As a result, dealers' option portfolios become more negatively gamma-exposed, leading them to hedge in the direction of price movements and amplify volatility in the underlying market. Consistent with this mechanism, we show that market makers' gamma declines most for stocks that experience larger increases in retail option trading, and that retail trades are more likely than non-retail trades to be intermediated by dealers.

The increase in volatility has important implications for market quality. Consistent with standard information models, greater noise trading improves liquidity—quoted and effective spreads and price impact decline—but worsens informational efficiency. In both the cross-listed and dual-class settings, price discovery shifts away from the markets most exposed to retail option trading. These findings suggest that the effects of retail option trading extend beyond trader-level outcomes and meaningfully alter the price-formation process in equity markets.

Our results highlight a previously underappreciated consequence of lowering barriers to retail option trading. Policies and platform designs that expand household access to derivative markets may improve trading opportunities for retail investors, but they can also generate externalities in the form of higher volatility and reduced price informativeness in the underlying securities. Accounting for these broader market effects is therefore essential in evaluating the costs and benefits of retail access to complex financial instruments.

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Figures

Figure 1
Predominance of Retail Trades in the Option Market

These figure shows the time-series evolution of 1-contract trades volume in option market, in absolute amounts and as a fraction of total volume. The sample is obtained from CBOE and consists of all option transactions that took place on US markets between January 2016 and December 2020. The vertical lines highlight the dates of December 13, 2017—the day Robinhood introduced option trading.

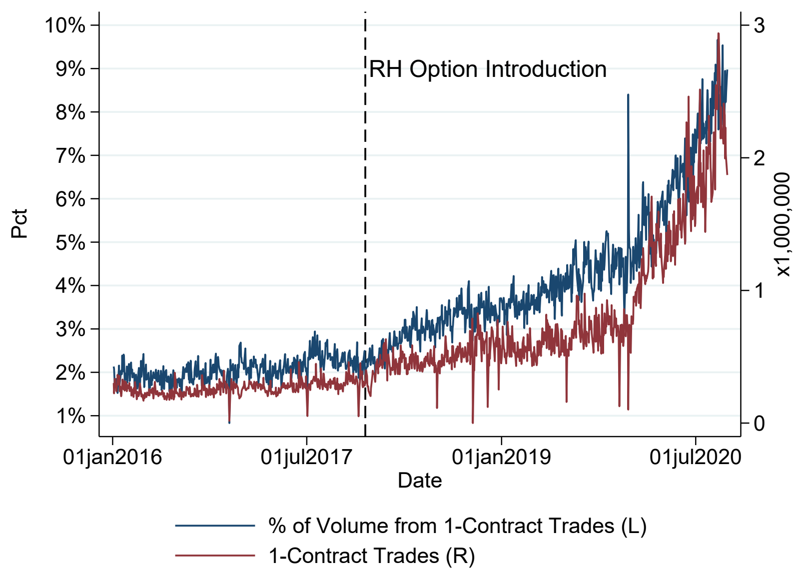


Figure 2
Increase in Option Retail Trading and Volatility Around Robinhood Free Option Introduction

This figure shows the time series evolution of option retail trading—measure as the number of 1-contract trades—around the introduction of free option trading on Robinhood, which took place on December 13, 2017. We also report the daily volatility for all stocks traded on US exchanges. Option retail trading is measured as number of 1-contract trades, in hundreds of thousands of trades. Volatility is based on 5-minutes returns. The sample extends from June 1, 2017, to May 31, 2018. Option trade data is obtained from CBOE, stock trade date comes from TAQ.

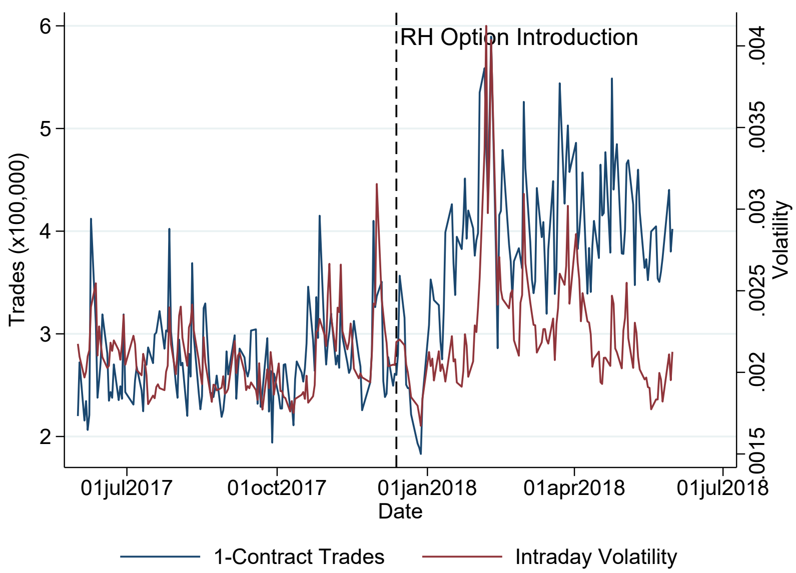


Figure 3
Google Searches and 1-Contract (Retail) Volume

This figure shows the relation between retail trading in options and Google searches volumes for terms related to option trading. We show the evolution of retail trading in options, measured as the number of 1-contract trades, and the Search Volume Index for the terms "Robinhood Options", "TD Ameritrade Options", "Schwab Options", and "Option Trading". The Search Volume Index is published by Google and measures the frequency with which terms are searched, normalized over the sample to vary between 0 and 100, where 100 is the maximum search intensity for the four series in the sample. The sample covers the 2016–2020 period. Option trade data is obtained from CBOE.

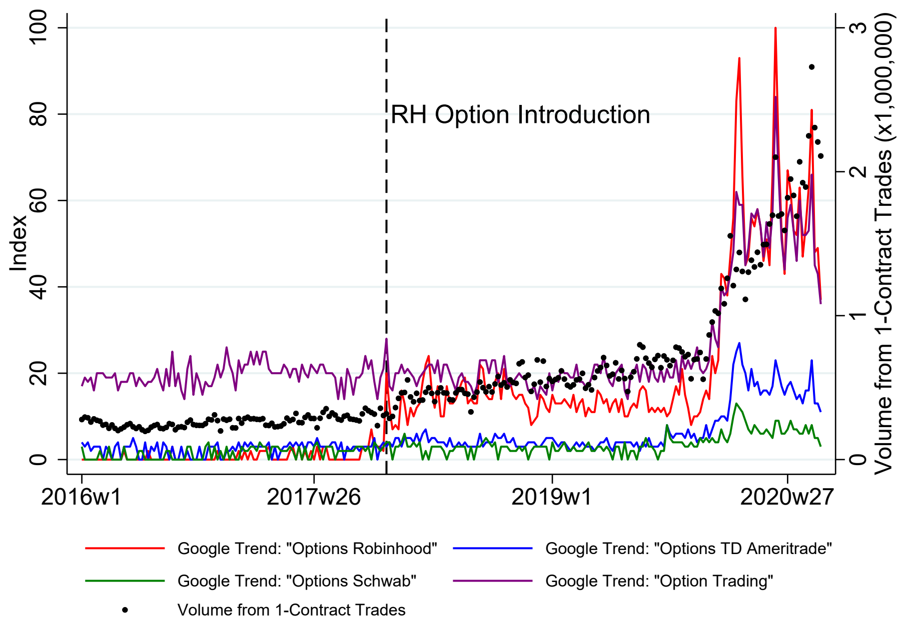
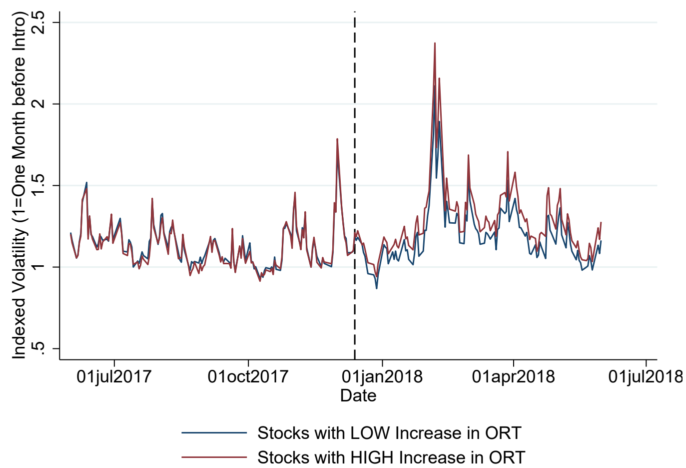


Figure 4
Stock Volatility and Option Retail Trading

This figure shows the stock market volatility for two groups of companies, based on whether they experienced a large or small increase in retail option trading around the introduction of free option trading on Robinhood. In Panel A, we show the volatility, standardized by dividing it by its value at the beginning of the sample. In Panel B, we shows the relative difference in volatility for the two groups of stocks, based on a difference-in-differences model, where the change in retail trading is the treatment and the post-period follows the introduction of options on Robinhood. The sample consists of daily observation for 3,359 stocks between June 1, 2017, and May 31, 2018.

A: Normalized volatility



B: Dynamic parameters

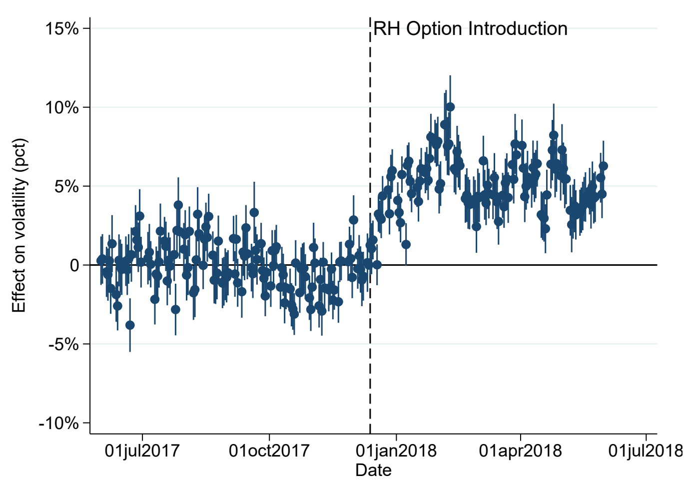
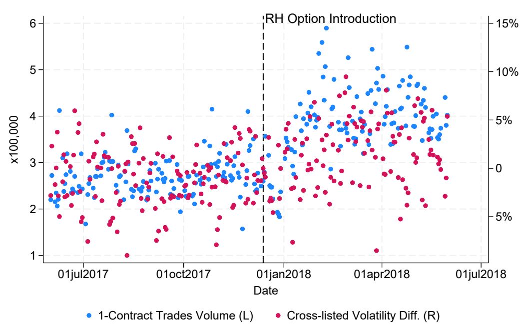


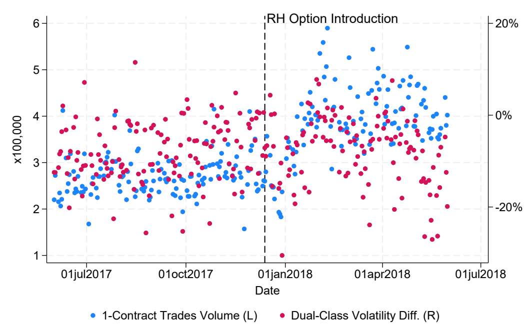
Figure 5
Retail Option Trading and Volatility: Non-parametric Difference-in-Differences

This figure shows the daily number of 1-contract option trades on CBOE (left y-axes), together with estimates of excess volatility in the stock market (right y-axis). The excess volatility is estimated as the average relative difference between the volatility of the US and Canadian shares of 71 interlisted companies, in Panel A, and as the average relative difference between the volatility of two share classes, for a set of 22 US companies for which two share classes are traded on the stock market, yet only one share class is the underlying of options, in Panel B. In Panel C, we measure excess volatility the absolute size and interquartile spread of the price deviation between the US and Canadian shares of interlisted companies. The vertical line marks the introduction of options on Robinhood. The sample spans June 1, 2017, to May 31, 2018. Option trade data is obtained from CBOE, stock prices data are obtained from TAQ and the Toronto Stock Exchange for US and Canadian shares, respectively.

A: Interlisted Counterfactual



B: Dual Class Counterfactual



C: Interlisted Price Deviation

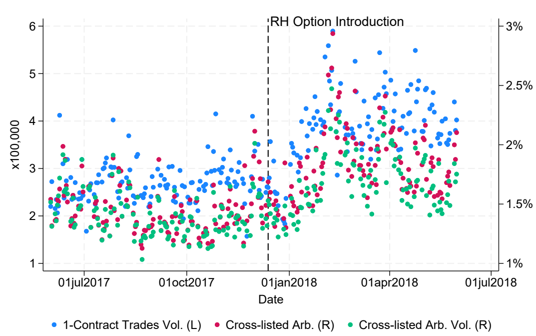
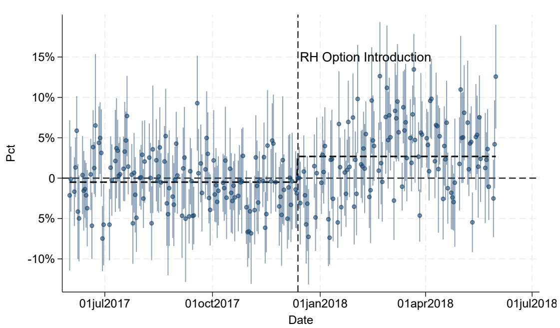


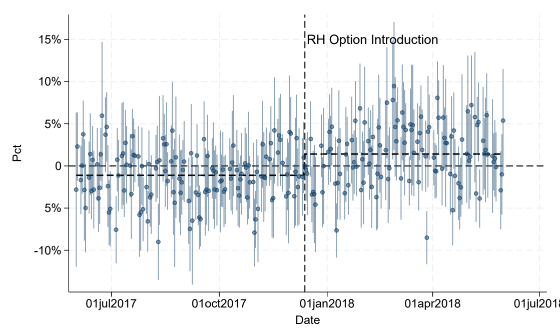
Figure 6
Option Retail Trading and Volatility: Difference-in-Differences

This figure shows the the difference-in-differences parameters for the specification in Eq. 4, and represents the relative difference in 5-minutes volatility between treated and non-treated shares. Panel A and B show the estimates for the increased volatility between interlisted US stocks and their Canadian counterparts. Panel A reports the difference in raw volatility, while Panel B replicates the analysis using USD-equivalent returns for the CAD-denominated shares. We estimate a diff-in-diff model, where the treated share is that traded in the US market, the control is its Canadian counterpart, and the post-period follows the introduction of free option trading on Robinhood. In Panel C we report the estimated diff-in-diff parameter for the setting where two share classes for the same company are traded on the stock market, yet only one share class is the underlying of options, constituting the treated sample. The sample spans June 1, 2017, to May 31, 2018. Stock prices data are obtained from TAQ and the Toronto Stock Exchange for US and Canadian shares, respectively.

A: Interlisted



B: Interlisted, ex-FX volatility



C: Dual class

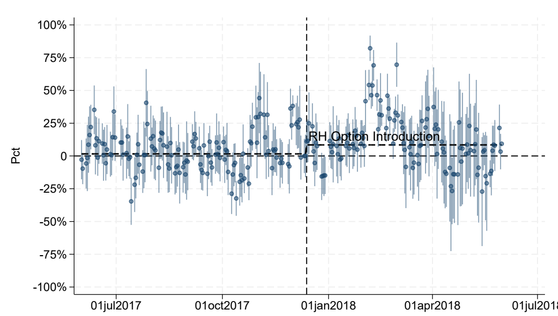


Figure 7
Delta and Gamma Demand for Retail and Non-Retail Trades

This figure shows the cumulative delta and gamma demand generated by option trade imbalances from small and large trades. Trade imbalances are computed at the option contract level separately for one-contract trades (retail) and larger trades (non-retail). Delta and gamma demand are obtained by multiplying contract-level imbalances by the option's delta and gamma, respectively, and aggregating to the stock–day level. The figure reports daily average quantities. Panels A–D reports the results for call option contracts, while Panels E–H reports the results for put option contracts. The vertical line marks Robinhood's introduction of free option trading. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts, aggregated to 420,689 observations at the day-stock level from June 1, 2017, to May 31, 2018.

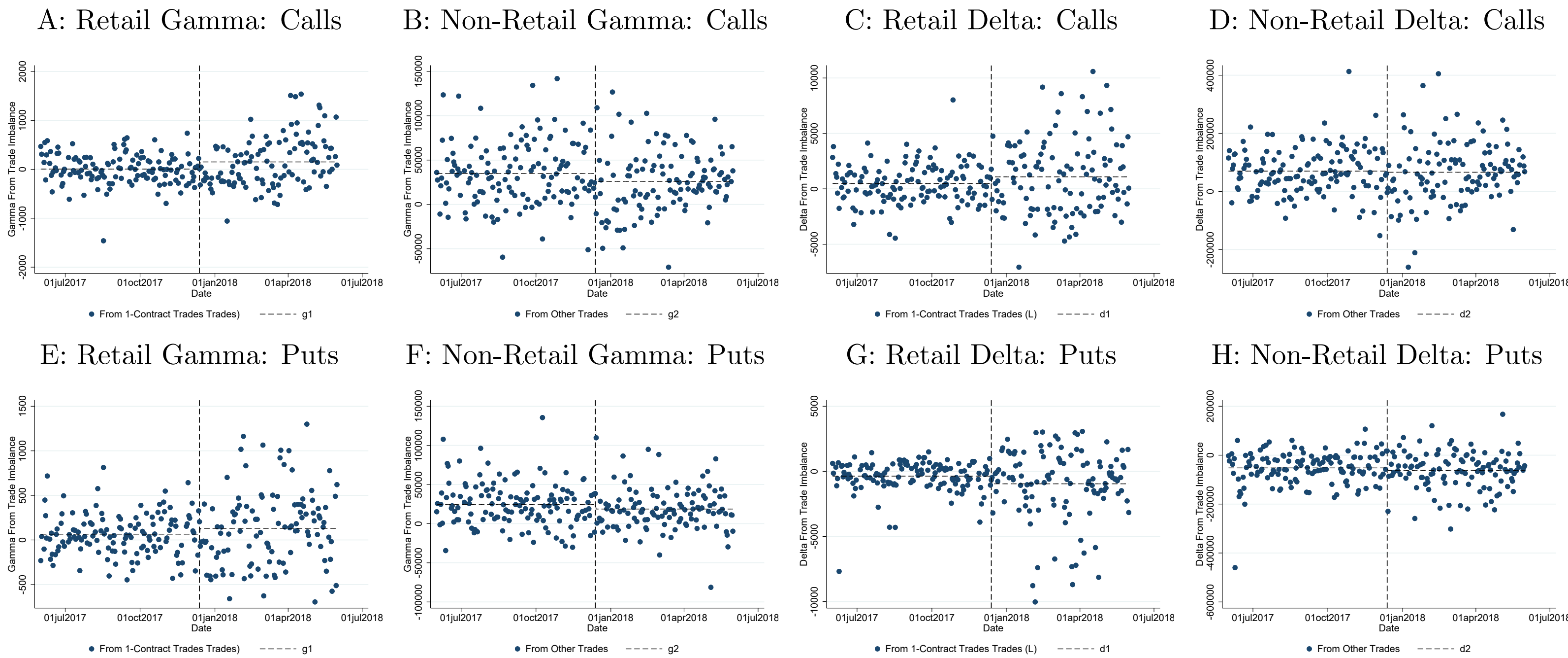
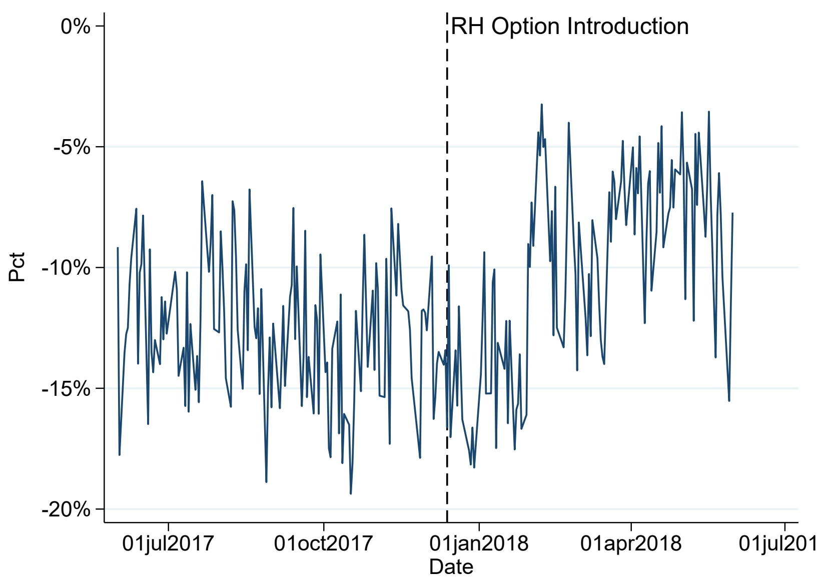


Figure 8
Cross-sectional Correlation between Retail Investing in the Stock and Option Market

This figure shows the cross-sectional (Spearman) rank-correlation between measures of retail trading in the stock and the option market. The measure for the option (stock) market is calculated as the fraction of trading volume coming from 1-contract trades (trades identified as retail as per the algorithm developed by Boehmer et al., 2021). The sample spans June 1, 2017, to May 31, 2018, option trade data is obtained from CBOE, stock prices data are obtained from TAQ.



Tables

Table 1
Volatility and Retail Option Trading

This table reports regressions of log five-minute stock price volatility on retail option trading volume. The dependent variable is log five-minute volatility, $Volatility_{it}$. The main explanatory variable is retail option trading volume measured by the number of one-contract option trades, $RetailOVolume_{it}$. Control variables include stock trading volume, $SVolume_{it}$; option trading volume, $OVolume_{it}$; and the quoted bid–ask spread, $QuotedSp_{it}$. All specifications include stock, date, and date–industry fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** indicate that parameters are significantly different from zero at the 10%, 5%, or 1% level, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

	(1) $Volatility_{it}$	(2) $Volatility_{it}$	(3) $Volatility_{it}$	(4) $Volatility_{it}$	(5) $Volatility_{it}$
$RetailOVolume_{it}$	0.107***	0.082***		0.033***	0.033***
	(45.417)	(43.310)		(23.726)	(23.726)
$OVolume_{it}$		0.035***	0.072***	0.013***	0.013***
		(45.472)	(45.967)	(25.047)	(24.914)
$RetailOVolume_{it}/OVolume_{it}$			0.193***		
			(39.284)		
$RetailSVolume_{it}$				0.064***	0.064***
				(27.328)	(28.039)
$SVolume_{it}$				0.177***	0.169***
				(37.460)	(36.339)
$QuotedSp_{it}$				0.972***	0.945***
				(34.059)	(33.899)
Stock FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes
Date-Industry FE	No	No	No	No	Yes
Adj. R^2	0.852	0.856	0.852	0.891	0.895
Obs	590,206	590,206	590,206	585,586	584,755

Table 2

Volatility and Retail Option Trading Around Robinhood's Option Introduction

This table reports difference-in-differences regressions examining changes in log five-minute stock price volatility around Robinhood's introduction of commission-free option trading. The dependent variable is $Volatility_{it}$. $IncreasedORT_i$ is an indicator equal to one for stocks experiencing a large increase in retail option trading following the fee reduction, and zero otherwise. $Post_t$ is an indicator equal to one for the post–option introduction period. Control variables include the quoted bid–ask spread, $QuotedSp_{it}$, and stock trading volume, $SVolume_{it}$. All specifications include stock, date, and date–industry fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

	(1) $Volatility_{it}$	(2) $Volatility_{it}$	(3) $Volatility_{it}$	(4) $Volatility_{it}$	(5) $Volatility_{it}$	(6) $Volatility_{it}$	(7) $RetailOVolume_{it}$	(8) $RetailOVolume_{it}$
$IncreasedORT_i{=}1 \times Post_t{=}1$		0.043^{***}	0.051^{***}	0.051^{***}	0.056^{***}	0.044^{***}		0.463^{***}
		(6.379)	(8.090)	(8.014)	(9.081)	(7.375)		(22.631)
$Post_t{=}1$	0.090^{***}	0.078^{***}	0.068^{***}				0.140^{***}	
	(5.463)	(4.703)	(4.142)				(9.663)	
$IncreasedORT_i{=}1$		-0.214^{***}						
		(−7.109)						
$QuotedSp_{it}$					0.517^{***}	0.489^{***}		-0.603^{***}
					(28.010)	(26.596)		(−12.588)
$SVolume_{it}$					0.126	0.087		0.229^{*}
					(1.511)	(1.288)		(1.719)
Stock FE	No	No	Yes	Yes	Yes	Yes	No	Yes
Date FE	No	No	No	Yes	Yes	Yes	No	Yes
Date-Ind. FE	No	No	No	No	No	Yes	No	Yes
Adj. R^2	0.003	0.016	0.823	0.847	0.855	0.871	0.001	0.850
Obs	810,971	810,971	810,971	810,971	810,797	750,422	590,207	531,938

Table 3
Descriptive Statistics

This table reports descriptive statistics for firms in the full sample. For each firm, we report market capitalization; daily stock trading volume; daily retail stock trading volume (in millions of dollars and as a fraction of total volume); daily option trading volume; daily retail option trading volume (in contracts and as a fraction of total volume); an indicator for S&P 500 membership; stock price; the price of a one-contract option trade; the average option traded price scaled by the stock price; implied and realized volatility; and option leverage (Omega), as defined in Section 2.1. For each variable, we calculate the median for each company in the period prior to Robinhood's option introduction, and we report in this table its average, calculated separately for firms experiencing an increase in retail option trading and for firms that do not. *, **, and *** indicate that the difference is significantly different from zero at the 10%, 5%, or 1% level, respectively. The sample consists of daily observation for 3,359 stocks between June 1, 2017, and May 31, 2018.

	$IncreasedORT_i = 0$	$IncreasedORT_i = 1$	Difference
Market Cap ($ bn)	8.500	8.339	−0.161
Volume in Underlying ($ bn)	0.050	0.054	0.004
Retail Volume in Underlying ($ mn)	4.142	4.465	0.323
Retail Volume in Underlying (pct)	0.092	0.109	0.017***
Volume in Option ('000 contracts)	2.537	3.960	1.423
Retail Volume in Option (contracts)	75.641	53.406	−22.235
Retail Volume in Option (pct)	0.041	0.043	0.002
SP500	0.145	0.113	−0.032**
Underlying Stock Price ($)	46.777	47.645	0.868
1-contract Option Price	167.372	155.783	−11.589**
Option-Stock Price Ratio	6.878	6.289	−0.590
Implied Volatility	0.407	0.371	−0.036***
Volatility (bps)	0.200	0.178	−0.022***
Omega	15.668	16.440	0.773

Table 4

Stock Volatility and Ex-Ante Exposure to Fee Reductions

This table reports regressions documenting the relation between volatility changes and ex-ante variables predicting retail volume increases after Robinhood's introduction of commission-free option trading. Panel A uses log five-minute stock price volatility, $Volatility_{it}$, as the dependent variable, while Panel B uses retail option trading volume measured by the number of one-contract option trades, $RetailOVolume_{it}$. $HighS_i$ is an indicator equal to one for stocks in the top tercile of stock prices over the three months prior to the fee reduction. $LowIV_i$ equals one if the volume-weighted average implied volatility of options written on the stock is in the bottom tercile. $LowO/S_{it}$ equals one for stocks with a low ratio of option prices to stock prices, and $HighOmega_{it}$ equals one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. $Post_t$ is an indicator equal to one for the post–option introduction period. All specifications include stock and date fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

Panel A: $Volatility_{it}$				
	(1)	(2)	(3)	(4)
$HighS_i$=1 × $Post_t$=1	0.073***			
	(9.218)			
$LowIV_i$=1 × $Post_t$=1		0.108***		
		(11.864)		
$LowO/S_i$=1 × $Post_t$=1			0.078***	
			(9.627)	
$HighOmega_i$=1 × $Post_t$=1				0.087***
				(11.020)
Stock FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Adj. R^2	0.848	0.848	0.848	0.848
Obs	810,971	810,971	810,971	810,971

Panel B: $RetailOVolume_{it}$				
	(1)	(2)	(3)	(4)
$HighS_i$=1 × $Post_t$=1	0.035*			
	(1.785)			
$LowIV_i$=1 × $Post_t$=1		0.096***		
		(4.673)		
$LowO/S_i$=1 × $Post_t$=1			0.047**	
			(2.377)	
$HighOmega_i$=1 × $Post_t$=1				0.081***
				(4.128)
Stock FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Adj. R^2	0.837	0.838	0.837	0.838
Obs	590,207	590,207	590,207	590,207

Table 5

Stock Volatility and Ex-Ante Exposure to Fee Reductions: One- and Two-way Sorts

This table reports changes in log five-minute stock price volatility, $Volatility_{it}$, and retail option trading volume measured by the number of one-contract option trades, $RetailOVolume_{it}$, around Robinhood's introduction of commission-free option trading. We report the changes based on pre-event quintile of ex-ante characteristics. For each stock, changes are computed as the difference between the post-event and pre-event periods. Panel A reports average changes by pre-event quintile of the ratio of average option prices to stock prices, O/S_{it}, and option leverage, defined as Omega= $\Delta S/O$. Panels B and C report analogous results sorting stocks by pre-event quintile of stock prices, S_i, and volume-weighted average implied volatility. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

Panel A: One-way sorts

	by O/S_i		by Omega	
Rank	$Volatility_{it}$	$RetailOVolume_{it}$	$Volatility_{it}$	$RetailOVolume_{it}$
0	0.348	0.334	0.022	0.172
1	0.197	0.231	0.082	0.202
2	0.140	0.211	0.133	0.211
3	0.076	0.208	0.218	0.248
4	0.027	0.211	0.335	0.346

Panel B: Two-way sorts, by IV_i (down) and S_i (across)
$Volatility_{it}$

	0	1	2	3	4
0	−1.000	0.059	0.451	0.481	0.465
1	−0.107	0.142	0.140	0.207	0.305
2	0.109	0.103	0.111	0.133	0.158
3	0.040	0.071	0.064	0.088	0.125
4	−0.001	0.009	0.018	0.037	0.097

Panel C: Two-way sorts, by IV_i (down) and S_i (across)
$RetailOVolume_{it}$

	0	1	2	3	4
0	0.349	0.213	0.426	0.411	0.385
1	1.067	0.378	0.257	0.235	0.227
2	0.500	0.218	0.207	0.173	0.173
3	0.343	0.238	0.269	−0.071	0.133
4	0.243	0.178	0.315	−0.066	0.036

Table 6

Stock Volatility at Varying Horizons and Ex-Ante Exposure to Fee Reductions

This table reports regressions examining changes in stock price volatility at multiple horizons around Robinhood's introduction of commission-free option trading. Daily volatility is measured using 5-, 10-, 30-, and 60-minute returns. Weekly and monthly volatility, $1WeekDayVol_{it}$ and $1MoDayVol_{it}$, are measured using open-to-close daily returns. $IncreasedORT_i$ is an indicator equal to one for stocks experiencing a large increase in retail option trading following the fee reduction. $HighS_i$ equals one for stocks in the top tercile of stock prices over the three months prior to the event. $LowIV_i$ equals one if the volume-weighted average implied volatility is in the bottom tercile. $LowO/S_{it}$ equals one for stocks with a low ratio of option prices to stock prices, and $HighOmega_{it}$ equals one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. $Post_t$ is an indicator equal to one for the post–option introduction period. All specifications include stock and time fixed effects. Standard errors are two-way clustered by stock and time-period. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

	$5MinVol_{it}$	$10MinVol_{it}$	$30MinVol_{it}$	$60MinVol_{it}$	$1WeekDayVol_{it}$	$1MoDayVol_{it}$
$IncreasedORT_i$=1 × $Post_t$=1	0.051***	0.055***	0.055***	0.058***	0.104***	0.122***
	(8.014)	(8.557)	(8.344)	(8.508)	(6.628)	(5.423)
Adj. R^2	0.847	0.829	0.770	0.692	0.605	0.804
Obs	810,971	810,968	810,957	810,951	171,968	39,022
	$5MinVol_{it}$	$10MinVol_{it}$	$30MinVol_{it}$	$60MinVol_{it}$	$1WeekDayVol_{it}$	$1MoDayVol_{it}$
$HighS_i$=1 × $Post_t$=1	0.073***	0.072***	0.077***	0.076***	0.167***	0.213***
	(9.218)	(8.934)	(9.082)	(8.376)	(5.048)	(4.796)
Adj. R^2	0.848	0.830	0.770	0.692	0.607	0.808
Obs	810,971	810,968	810,957	810,951	171,968	39,022
	$5MinVol_{it}$	$10MinVol_{it}$	$30MinVol_{it}$	$60MinVol_{it}$	$1WeekDayVol_{it}$	$1MoDayVol_{it}$
$LowIV_i$=1 × $Post_t$=1	0.108***	0.109***	0.114***	0.119***	0.210***	0.260***
	(11.864)	(11.619)	(11.234)	(11.059)	(5.206)	(4.137)
Adj. R^2	0.848	0.830	0.771	0.692	0.608	0.810
Obs	810,971	810,968	810,957	810,951	171,968	39,022
	$5MinVol_{it}$	$10MinVol_{it}$	$30MinVol_{it}$	$60MinVol_{it}$	$1WeekDayVol_{it}$	$1MoDayVol_{it}$
$LowO/S_i$=1 × $Post_t$=1	0.078***	0.079***	0.083***	0.085***	0.152***	0.193***
	(9.627)	(9.440)	(9.290)	(9.043)	(4.555)	(4.317)
Adj. R^2	0.848	0.830	0.770	0.692	0.606	0.807
Obs	810,971	810,968	810,957	810,951	171,968	39,022
	$5MinVol\_it$	$10MinVol\_it$	$30MinVol\_it$	$60MinVol\_it$	$1WeekDayVol\_it$	$1MoDayVol\_it$
$HighOmega\_i$=1 × $Post\_t$=1	0.087***	0.088***	0.094***	0.095***	0.162***	0.210***
	(11.020)	(10.846)	(11.024)	(10.463)	(5.143)	(4.525)
Adj. R^2	0.848	0.830	0.771	0.692	0.606	0.807
Obs	810,971	810,968	810,957	810,951	171,968	39,022

Table 7

Retail Option Trading and Stock Volatility: Difference-in-Differences Evidence from Cross-Listed Canadian Firms

This table reports difference-in-differences regressions examining changes in log five-minute stock price volatility around Robinhood's introduction of commission-free option trading. Identification exploits differential exposure to commission-free option trading across otherwise identical cross-listed shares. The sample consists of 71 Canadian firms with shares cross-listed in the United States and Canada. The dependent variable is log five-minute volatility, $Volatility_{it}$. US_i is an indicator equal to one for the U.S.-dollar–denominated share and zero for the Canadian-dollar–denominated share. In Specifications 3 and 4 of Panel A, the dependent variable is the log difference in volatility between the U.S.- and Canadian-listed shares of the same firm, $Vol_{it}^{US} - Vol_{it}^{CA}$, or the corresponding difference adjusted for foreign-exchange volatility, $Vol_{it}^{US} - Vol_{it}^{CA,USD}$, respectively. $Post_t$ is an indicator equal to one for the post–option introduction reduction period. In Panel B, the dependent variables are the interquartile spread of the price deviation between U.S.- and Canadian-listed shares, $Vol_{it}^{Dev.}$, and the average absolute price difference between the two shares, $Deviation_{it}$. All specifications include stock and date fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period spans June 1, 2017, to May 31, 2018.

Panel A: Volatility

	(1) $Volatility_{it}$	(2) $Volatility_{it}$	(3) $Vol_{it}^{US} - Vol_{it}^{CA}$	(4) $Vol_{it}^{US} - Vol_{it}^{CA,USD}$
US_i=1 × $Post_t$=1	0.031***	0.024**		
	(2.904)	(2.453)		
$Post_t$=1			0.024**	0.025**
			(2.209)	(2.584)
$FxVol_t$			0.059***	
			(4.306)	
Stock FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	No	No
Adj. R^2	0.778	0.776	0.151	0.161
Obs	33,033	33,035	16,488	16,490

Panel B: Absolute Price Deviation

	(1) $Vol_{it}^{Dev.}$	(2) $Vol_{it}^{Dev.}$	(3) $Deviation_{it}$
$Post_t$=1	0.003***	0.003***	0.004***
	(3.707)	(3.369)	(3.603)
$FxVol_t$		0.003***	0.003***
		(2.699)	(3.407)
Stock FE	Yes	Yes	Yes
Time FE	No	No	No
Adj. R^2	0.734	0.735	0.750
Obs	16,545	16,545	16,545

Table 8

Retail Option Trading and Stock Volatility: Triple-Differences Evidence from Cross-Listed Canadian Firms

This table reports difference-in-differences-in-differences regressions examining changes in log five-minute stock price volatility around Robinhood's introduction of commission-free option trading. The sample consists of 71 Canadian firms with shares cross-listed in the United States and Canada. The dependent variable is log five-minute volatility, $Volatility_{it}$. $Treated_i$ is an indicator equal to one for the U.S.-dollar–denominated share and zero for the Canadian-dollar–denominated share. $Post_t$ is an indicator equal to one for the post–option introduction period. $HighS_i$ equals one if the U.S.-listed share is in the top tercile of stock prices over the three months prior to the event. $LowIV_i$ equals one if the volume-weighted average implied volatility is in the bottom tercile. $LowO/S_{it}$ equals one for stocks with a low ratio of option prices to stock prices, and $HighOmega_{it}$ equals one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. All specifications include stock fixed effects, stock-by-period fixed effects, and treated-status–by-period fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period spans June 1, 2017, to May 31, 2018.

	(1) $Volatility_{it}$	(2) $Volatility_{it}$	(3) $Volatility_{it}$	(4) $Volatility_{it}$
US_i=1 × $Post_t$=1 × $HighS_i$=1	0.071***			
	(3.709)			
US_i=1 × $Post_t$=1 × $LowIV_i$=1		0.040**		
		(2.043)		
US_i=1 × $Post_t$=1 × $LowO/S_i$=1			0.052***	
			(2.664)	
US_i=1 × $Post_t$=1 × $HighOmega_i$=1				0.039*
				(1.902)
Stock FE	Yes	Yes	Yes	Yes
Time-Treated FE	Yes	Yes	Yes	Yes
Stock-Post FE	Yes	Yes	Yes	Yes
Adj. R^2	0.786	0.786	0.786	0.786
Obs	33,035	33,035	33,035	33,035

Table 9

Retail Option Trading, Stock Volatility, and Liquidity: Difference-in-Differences Evidence from Dual-Class Shares

This table reports difference-in-differences regressions examining changes in log five-minute stock price volatility and liquidity around Robinhood's introduction of commission-free option trading. The sample consists of 22 U.S. firms with multiple share classes, at least one of which underlies traded option contracts. The dependent variables include log five-minute volatility, $Volatility_{it}$; the relative quoted, effective, and realized bid–ask spreads, $QuotedSp_{it}$, $EffectiveSp_{it}$, and $RealizedSp_{it}$; and a five-minute measure of price impact, $PriceImpact_{it}$. $Treated_i$ is an indicator equal to one for share classes underlying option contracts and zero otherwise. $Post_t$ is an indicator equal to one for the post–fee reduction period. All specifications include share-class and date fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period spans June 1, 2017, to May 31, 2018.

	(1) $Volatility_{it}$	(2) $QuotedSp_{it}$	(3) $EffectiveSp_{it}$	(4) $RealizedSp_{it}$	(5) $PriceImpact_{it}$
$Optioned_i{=}1 \times Post_t{=}1$	0.030**	−0.056**	−0.079***	−0.046***	−0.013**
	(2.022)	(−2.464)	(−6.179)	(−3.562)	(−2.048)
Stock FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.467	0.884	0.829	0.717	0.487
Obs	6,981	6,981	6,311	6,304	6,304

Table 10

Option Market Making and Ex-Ante Predictors of Retail Option Volume Increases

This table reports regressions examining changes in option market making around Robinhood's introduction of commission-free option trading. Panel A uses an indicator for positive market maker net gamma, $NegativeNetGamma_{it}$, which equals one if market makers' net gamma for stock i on day t is negative and zero otherwise. Panel B uses market makers' net gamma, $NetGamma_{it}$. $HighS_i$ is an indicator equal to one for stocks in the top tercile of stock prices over the three months prior to the fee reduction. $LowIV_i$ equals one if the volume-weighted average implied volatility is in the bottom tercile. $LowO/S_{it}$ equals one for stocks with a low ratio of option prices to stock prices, and $HighOmega_{it}$ equals one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. $Post_t$ is an indicator equal to one for the post–option introduction period. All specifications include stock and date fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

Panel A: Negative Market Maker's Net Gamma				
	(1)	(2)	(3)	(4)
$HighS_i$=1 × $Post_t$=1	0.019**			
	(2.231)			
$LowIV_i$=1 × $Post_t$=1		0.030***		
		(3.367)		
$LowO/S_i$=1 × $Post_t$=1			0.031***	
			(3.499)	
$HighOmega_i$=1 × $Post_t$=1				0.029***
				(3.237)
Stock FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Adj. R^2	0.125	0.125	0.125	0.125
Obs	692,270	692,270	692,270	692,270

Panel B: Market Maker's Net Gamma				
	(1)	(2)	(3)	(4)
$HighS_i$=1 × $Post_t$=1	−11.498***			
	(−3.606)			
$LowIV_i$=1 × $Post_t$=1		−7.342**		
		(−2.327)		
$LowO/S_i$=1 × $Post_t$=1			−10.425***	
			(−3.202)	
$HighOmega_i$=1 × $Post_t$=1				−8.666***
				(−2.744)
Stock FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Adj. R^2	0.176	0.176	0.176	0.176
Obs	692,270	692,270	692,270	692,270

Table 11
Market Maker Inventories and Trade Imbalances

This table reports regressions examining the relation between market maker inventories and trade imbalances. The dependent variable for Specification 1 (2) is the number of option contracts bought (sold) by market maker for options with stock-i as underlying on day-t, $MMBuy_{it}$ ($MMSell_{it}$). The dependent variable for Specification 3 is the change in market makers' net gamma, D.$NetGammaMM_{it}$. The explanatory variables measure the number of (retail) call or put contracts bought or sold by investors, or their imbalance, also aggregated at the stock-i, day-t level. All specifications include stock and date fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

	(1) $MMBuy_{it}$	(2) $MMSell_{it}$	(3) D.$Net\Gamma MM_{it}$
$TotalCallSell_{it}$	0.085***		
	(32.168)		
$RetailCallSell_{it}$	1.732***		
	(21.298)		
$TotalPutSell_{it}$	0.087***		
	(30.046)		
$RetailPutSell_{it}$	2.019***		
	(19.587)		
$TotalCallBuy_{it}$		0.093***	
		(36.276)	
$RetailCallBuy_{it}$		1.629***	
		(21.054)	
$TotalPutBuy_{it}$		0.096***	
		(32.149)	
$RetailPutBuy_{it}$		1.986***	
		(19.645)	
$TotalImbalanceCall_{it}$			−0.006***
			(−11.999)
$RetailImbalanceCall_{it}$			−0.117***
			(−5.605)
$TotalImbalancePut_{it}$			−0.007***
			(−11.153)
$RetailImbalancePut_{it}$			−0.094***
			(−3.097)
Stock FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Adj. R^2	0.907	0.904	−0.003
Obs	458,356	458,356	539,100

Table 12

Buy–Sell Activity and Imbalance in Option Contracts

This table reports contract–day averages of traded option quantities. For each contract–day, we report average buy and sell volumes separately for retail and non-retail trades, as well as the average absolute buy–sell imbalance. Contract-days observations are sorted into deciles based on total imbalance, defined as the difference between buy and sell volume at the contract level. Within each decile, the table reports also the correlation between retail buy and sell quantities, the correlation between non-retail buy and sell quantities, and the slope coefficient from regressions of buy quantities on sell quantities for retail and non-retail trades. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts from June 1, 2017, to May 31, 2018.

	1	2	3	4	5	6	7	8	9	10
Retail Buy	0.688	0.938	1.199	1.339	1.388	2.050	2.750	3.698	5.781	14.312
Retail Sell	0.715	0.967	1.231	1.380	1.428	2.108	2.823	3.768	5.813	13.821
Non-Retail Buy	3.482	5.528	8.031	10.957	13.689	23.004	36.579	61.467	131.139	1121.485
Non-Retail Sell	3.478	5.581	8.124	11.159	14.197	23.819	37.894	63.121	131.671	1036.894
Imbalance	0.559	2.000	3.449	5.787	9.528	15.846	26.381	46.459	101.288	1127.005
Corr(RetailBuy,RetailSell)	0.712	0.745	0.733	0.751	0.768	0.777	0.785	0.809	0.827	0.762
Corr(NonRetailBuy,NonRetailSell)	1.000	0.999	0.998	0.997	0.993	0.989	0.981	0.969	0.949	0.418
Beta(RetailBuy,RetailSell)	0.710	0.741	0.720	0.728	0.762	0.774	0.766	0.804	0.816	0.768
Beta(NonRetailBuy,NonRetailSell)	0.999	0.999	0.998	0.998	0.993	0.990	0.984	0.971	0.951	0.452

Table 13

Retail Option Trading and Contracts' Gamma

This table reports regressions examining the relation between retail option trading and option contract characteristics. The dependent variables are the relative trade imbalance of one-contract trades $RetailImb\%_{it}$ in Specification 1, the relative trade imbalance of larger trades $NonRetailImb\%_{it}$ in Specification 2, and the difference between the two measures in Specification 3, all measured at the contract–day level. As explanatory variables, we include the option price, $Price_{it}$, and gamma, $Gamma_{it}$. Specifications 4–6 replicate Specifications 1–3 and include the interaction between $Gamma_{it}$ and $Post_t$, an indicator equal to one for the period after Robinhood's option introduction. All specifications include underlying stock–by–day fixed effects. Standard errors are two-way clustered by contract and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts from June 1, 2017, to May 31, 2018.

	(1)	(2)	(3)	(4)	(5)	(6)
	$RetailImb\%_{it}$	$NonRetailImb\%_{it}$	$RetailImb\%_{it}-$ $NonRetailImb\%_{it}$	$RetailImb\%_{it}$	$NonRetailImb\%_{it}$	$RetailImb\%_{it}-$ $NonRetailImb\%_{it}$
$Gamma_{it}$	0.022***	0.012**	0.020***	0.009	0.015**	0.005
	(2.903)	(2.002)	(3.487)	(0.970)	(2.056)	(0.693)
$Post_t{=}1 \times Gamma_{it}$				0.027*	−0.007	0.033***
				(1.794)	(−0.609)	(2.865)
$Price_{it}$	0.020***	0.033***	−0.005	0.020***	0.033***	−0.005
	(4.934)	(7.576)	(−1.534)	(4.939)	(7.574)	(−1.522)
FE	$S \times D$	$S \times D$	$S \times D$	$S \times D$	$S \times D$	$S \times D$
Adj. R^2	0.027	0.024	0.011	0.027	0.024	0.011
Obs	7,350,048	8,910,580	6,277,306	7,350,048	8,910,580	6,277,306

Table 14

Changes in Retail and Non-Retail Imbalance by Level of Gamma

This table reports average trade imbalances, computed at the option contract-day level separately for one-contract trades (retail) and larger trades (non-retail). For each stock-day, options are sorted into quintiles based on their respective gamma. The table reports average imbalances for each quintile, separately for the periods before and after Robinhood's introduction of option trading, as well as the difference between the two periods. Panel A reports the results for retail trades, while Panel B reports the results for non-retail trades. *, **, and *** denote statistical significance of the difference at the 10%, 5%, and 1% levels, respectively. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts, from June 1, 2017, to May 31, 2018. May 31, 2018.

Panel A: Retail Imbalance			
Quintile	Before	After	Difference
Q1	0.066	0.076	0.011
Q2	0.005	−0.041	−0.045***
Q3	−0.039	−0.056	−0.017
Q4	−0.061	−0.052	0.009
Q5	0.063	0.140	0.077***

Panel B: Non-Retail Imbalance			
Quintile	Before	After	Difference
Q1	7.543	5.232	−2.311*
Q2	3.164	0.232	−2.932**
Q3	3.792	2.403	−1.390
Q4	8.402	5.774	−2.628
Q5	22.418	21.242	−1.175

Table 15

Delta and Gamma Demand from Retail and Non-Retail Option Trades

This table reports cumulative delta and gamma demand generated by option trade imbalances from small and large trades. Trade imbalances are computed at the option contract-day level separately for one-contract trades (retail) and larger trades (non-retail), then aggregated at the stock-day level. Delta and gamma demand are obtained by multiplying contract-level imbalances by the option's delta and gamma, respectively, and aggregating to the stock–day level. The table reports average quantities for the periods before and after Robinhood's introduction of option trading, as well as the difference between the two periods. Panel A reports the results for call option contracts, while Panel B reports the results for put option contracts. *, **, and *** denote statistical significance of the difference at the 10%, 5%, and 1% levels, respectively. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts, aggregated to 463,261 observations at the day-stock level from June 1, 2017, to May 31, 2018.

Panel A: Call Contracts

	$Before$	$After$	Difference
$ImbalanceFromRetail_{it}$	−0.312	−0.275	0.036
$ImbalanceFromNonRetail_{it}$	115.009	92.948	−22.061*
$GammaFromRetail_{it}$	0.003	0.087	0.084***
$GammaFromNonRetail_{it}$	21.400	14.954	−6.447***
$DeltaFromRetail_{it}$	0.291	0.622	0.330***
$DeltaFromNonRetail_{it}$	42.761	38.062	−4.699

Panel B: Put Contracts

	$Before$	$After$	Difference
$ImbalanceFromRetail_{it}$	0.560	0.754	0.194
$ImbalanceFromNonRetail_{it}$	120.908	113.482	−7.426
$GammaFromRetail_{it}$	0.049	0.091	0.042**
$GammaFromNonRetail_{it}$	18.431	13.067	−5.364**
$DeltaFromRetail_{it}$	−0.276	−0.668	−0.392***
$DeltaFromNonRetail_{it}$	−39.340	−43.544	−4.204

Table 16
Retail Option Trading and Contract Characteristics

This table reports regressions examining the relation between retail option trading and option contract characteristics. The dependent variable is the fraction of trading volume attributable to one-contract trades, $Retail\%_{it}$, measured at the contract–day level. Explanatory variables include the option price, $Price_{it}$, the strike price, $Strike_{it}$, and time to expiration, $TimeToExp_{it}$. Specifications (1) and (2) include underlying stock–by–expiration date–by–day fixed effects and restrict the sample to call options. Specifications (3) and (4) include underlying stock–by–strike price–by–day fixed effects and restrict the sample to put options. Standard errors are two-way clustered by contract and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts from June 1, 2017, to May 31, 2018.

	(1) $Retail\%_{it}$	(2) $Retail\%_{it}$	(3) $Retail\%_{it}$	(4) $Retail\%_{it}$
$Strike_{it}$	0.031***	−0.015***		
	(12.361)	(−8.725)		
$TimeToExp_{it}$			0.056***	0.050***
			(50.087)	(27.907)
$Price_{it}$	0.173***	0.174***	0.222***	0.373***
	(22.704)	(21.968)	(23.963)	(27.858)
FE	$S \times D \times T$	$S \times D \times T$	$S \times D \times K$	$S \times D \times K$
Put/Call Sample	C	P	C	P
Adj. R^2	0.199	0.226	0.193	0.176
Obs	5,039,530	3,983,748	4,376,892	3,339,073

Table 17

Retail Option Trading and Stock Liquidity: Difference-in-Differences Evidence from Cross-Listed Canadian Firms

This table reports difference-in-differences regressions examining changes in liquidity measures around Robinhood's introduction of commission-free option trading. Identification exploits differential exposure to commission-free option trading across otherwise identical cross-listed shares. The sample consists of 71 Canadian firms with shares cross-listed in the United States and Canada. The dependent variables are the relative quoted, effective, and realized bid–ask spreads, $QuotedSp_{it}$, $EffectiveSp_{it}$, and $RealizedSp_{it}$; and a five-minute measure of price impact, $PriceImpact_{it}$. US_i is an indicator equal to one for the U.S.-dollar–denominated share and zero for the Canadian-dollar–denominated share. All specifications include stock and date fixed effects. Standard errors are clustered at the date level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period spans June 1, 2017, to May 31, 2018.

	(1) $QuotedSp_{it}$	(2) $EffectiveSp_{it}$	(3) $RealizedSp_{it}$	(4) $PriceImpact_{it}$
$US_i{=}1 \times Post_t{=}1$	-0.005^{***}	-0.007^{***}	0.001	-0.007^{***}
	(-4.430)	(-6.532)	(0.684)	(-3.803)
Stock FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Adj. R^2	0.908	0.876	0.315	0.640
Obs	31,474	31,474	31,474	31,474

Table 18

Information Share of Canadian Cross-Listed Stocks around Robinhood Option Introduction

This table reports regressions examining changes in the information share of Canadian cross-listed stocks around Robinhood's introduction of commission-free option trading. The dependent variable is the information share of the Canadian market, $InfoShareCA_{it}$, defined as the fraction of the variance of the implicit efficient price attributable to innovations in Canadian quotes. Information shares are computed following Hasbrouck (1995) using a vector autoregression of U.S. and Canadian midquotes sampled at five-minute intervals. Estimates are constructed as the average of the upper and lower bounds. The sample consists of 71 Canadian firms with shares cross-listed in the United States and Canada. $HighS_i$ is an indicator equal to one if the U.S.-listed share is in the top tercile of stock prices over the three months prior to the fee reduction. $LowIV_i$ equals one if the volume-weighted average implied volatility is in the bottom tercile. $LowO/S_{it}$ equals one for stocks with a low ratio of option prices to stock prices, and $HighOmega_{it}$ equals one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. $Post_t$ is an indicator equal to one for the post–option introduction period. All specifications include stock fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period spans June 1, 2017, to May 31, 2018.

	(1) $InfoShareCA_{it}$	(2) $InfoShareCA_{it}$	(3) $InfoShareCA_{it}$	(4) $InfoShareCA_{it}$	(5) $InfoShareCA_{it}$
$Post_t$=1	0.036***	0.015**	0.027***	0.022***	0.028***
	(4.518)	(2.065)	(3.100)	(2.872)	(3.327)
$HighS_i$=1 × $Post_t$=1		0.071***			
		(4.685)			
$LowIV_i$=1 × $Post_t$=1			0.030*		
			(1.834)		
$LowO/S_i$=1 × $Post_t$=1				0.045***	
				(2.694)	
$HighOmega_i$=1 × $Post_t$=1					0.031*
					(1.769)
Stock FE	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.395	0.404	0.397	0.399	0.397
Obs	16,911	16,911	16,911	16,911	16,911

Table 19

Information Share of Dual-Class Shares around Robinhood Option Introduction

This table reports regressions examining changes in the information share of non-optioned shares around Robinhood's introduction of commission-free option trading. The dependent variable is the information share of the non-optioned share class, $InfoShareNonOptioned_{it}$, defined as the fraction of the variance of the implicit efficient price attributable to innovations in quotes for the non-optioned share. Information shares are computed following Hasbrouck (1995) using a vector autoregression of midquotes for optioned and non-optioned share classes sampled at five-minute intervals. Estimates are constructed as the average of the upper and lower bounds. The sample consists of firms with multiple share classes, some of which underlie traded option contracts. $Post_t$ is an indicator equal to one for the post–fee reduction period. All specifications include firm fixed effects. Standard errors are clustered by date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample period spans June 1, 2017, to May 31, 2018 for Specification 1, and January 1, 2017, to December 31, 2018 for Specification 2.

	(1) $InfoShareNotOptioned_{it}$	(2) $InfoShareNotOptioned_{it}$
$Post_t$	0.011*	0.040***
	(1.720)	(8.936)
Stock FE	Yes	Yes
Adj. R^2	0.261	0.217
Obs	6,848	13,389

Appendix

A.1 A Comparison Between Option Retail Trading Measures

In the main body of the paper, we proxy for retail option trading using the fraction of trading volume accounted for by one-contract trades. Bryzgalova et al. (2023) develop an institutionally grounded measure that identifies retail option trades based on detailed trade characteristics. In this section, we assess how closely our proxy aligns with the measures proposed by Bryzgalova et al. (2023).

Retail option orders are typically routed by brokers—e.g., Robinhood, TD Ameritrade, or Charles Schwab—to wholesalers—e.g., Citadel, Susquehanna, Wolverine—in exchange for payment for order flow (PFOF). Unlike in equity markets, where wholesalers may execute trades internally on their platforms, retail option orders must ultimately be executed on an exchange. Wholesalers can nevertheless ensure that they be the counterparty to the trade by posting the order on the market and submitting it through a price-improvement auction, typically via an affiliated market maker. Other market participants may compete to improve the price, but exchange fee structures make it prohibitively costly to break up the trade. As a result, trades executed through price-improvement mechanisms provide a reliable proxy for retail option activity. In Options Price Reporting Authority (OPRA) data, such trades are flagged as "SLAN" (single-leg price improvement mechanism).

Bryzgalova et al. (2023) further propose an expanded measure of retail trading based on an alternative proxy for internalization. When a market maker is designated as a primary market maker in a given option and is quoting at the national best bid and ask, it has execution priority for trades up to five contracts. These executions effectively constitute internalized trades. The authors capture this activity using trades flagged as single-leg electronic orders with an "AUTO" designation. The combined volume of SLAN- and AUTO-flagged trades forms their "All Internalized" measure.

The flags underlying the SLAN and AUTO measures become available only in November 2019, subsequently to the sample period of our main analysis. However, for the overlapping period from November 2019 through September 2020, when both our data and the measures made available by Bryzgalova et al. (2023) are observable, we can verify that the proxies are generally aligned.

We aggregate the one-contract measure, the SLAN measure, and the All Internalized measure at the ticker-day level by summing retail trading activity across all option contracts, which results in 623,701 observations. Figure A-1 presents binscatter plots of the joint distributions of our one-contract proxy and the two measures proposed by Bryzgalova et al. (2023), and show that the measures closely track each other. The one-contract proxy's rank correlation with the SLAN measure is 84%, and 90% with the All Internalized measure (Panels A and C). To account for heterogeneity in overall option trading volume, we rescale each measure by total option volume at the ticker-day level; the corresponding rank correlations are 4% and 38% (Panels B and D), respectively.

To ensure that these relationships are not driven solely by cross-sectional or time-series variation, we repeat the analysis after orthogonalizing all measures with respect to ticker and date fixed effects. Figure A-2 reports the resulting comparisons. The strong alignment persists, indicating that the proxy used in this paper closely tracks the retail trading measures developed by Bryzgalova et al. (2023) along both dimensions.

Table A-1

Heterogeneity in the Effects of Commission-Free Option Trading: Regressions

This table reports regressions examining heterogeneity in the effects of Robinhood's introduction of commission-free option trading. Panels A and B use log five-minute stock price volatility, $Volatility_{it}$, as the dependent variable, Panel C uses retail option trading volume measured by the number of one-contract option trades, $RetailOVolume_{it}$, and Panel D reports the number of stocks in each two-by-two quantile of stock price and implied volatility. Each panel reports coefficients from regressions that include indicator variables for a stock's rank in the relevant characteristic interacted with $Post_t$, an indicator equal to one for the post–option introduction period. In Panel A, quintiles are formed based on pre-event values of the option price–to–stock price ratio and option leverage. In Panels B and C, regressions are estimated separately by pre-event quintiles of stock price, with explanatory variables indicating pre-event quintiles of implied volatility. All specifications include stock and date fixed effects. Standard errors are two-way clustered by stock and date. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The sample consists of daily observations for 3,359 stocks from June 1, 2017, to May 31, 2018.

Panel A: One-way sorts

	by O/S_i		by $\frac{\Delta \cdot S}{O}$	
	$Volatility_{it}$	$RetailOVolume_{it}$	$Volatility_{it}$	$RetailOVolume_{it}$
$Rank_i$=0 × $Post_t$=1	0.116***	0.110***	−0.128***	−0.143***
$Rank_i$=1 × $Post_t$=1	0.075***	0.045	−0.102***	−0.081**
$Rank_i$=2 × $Post_t$=1	0.051***	0.026	−0.081***	−0.080***
$Rank_i$=3 × $Post_t$=1	0.018	0.034	−0.046***	−0.046*
$Rank_i$=4 × $Post_t$=1				

Panel B: Two-way sorts, by IV_i (down) and S_i (across)
$Volatility_{it}$

	$RankS_i$=0	$RankS_i$=1	$RankS_i$=2	$RankS_i$=3	$RankS_i$=4
$RankIV_i$=0 × $Post_t$=1	−0.350	0.118***	0.155***	0.205***	0.176***
$RankIV_i$=1 × $Post_t$=1	−0.025	0.134***	0.019	0.116***	0.140***
$RankIV_i$=2 × $Post_t$=1	0.081*	0.097***	0.011	0.072**	0.073**
$RankIV_i$=3 × $Post_t$=1	0.016	0.084***	−0.016	0.042	0.097**
$RankIV_i$=4 × $Post_t$=1					

Panel C: Two-way sorts, by IV_i (down) and S_i (across)
$RetailOVolume_{it}$

	$RankS_i$=0	$RankS_i$=1	$RankS_i$=2	$RankS_i$=3	$RankS_i$=4
$RankIV_i$=0 × $Post_t$=1	0.485***	0.064	−0.059	0.229*	0.330**
$RankIV_i$=1 × $Post_t$=1	−0.005	0.256***	−0.087	0.132	0.254*
$RankIV_i$=2 × $Post_t$=1	0.170	0.047	−0.060	0.067	0.208
$RankIV_i$=3 × $Post_t$=1	0.023	0.060	−0.142	−0.009	0.273*
$RankIV_i$=4 × $Post_t$=1					

Panel D: Two-way sorts, by IV_i (down) and S_i (across)
Frequency

	$RankS_i$=0	$RankS_i$=1	$RankS_i$=2	$RankS_i$=3	$RankS_i$=4
0	3.000	51.000	112.000	171.000	281.000
1	6.000	85.000	123.000	200.000	208.000
2	21.000	135.000	166.000	174.000	126.000
3	109.000	244.000	145.000	79.000	46.000
4	424.000	114.000	47.000	20.000	13.000

Table A-2

Bootstrapped Distribution of Difference-in-Differences Estimates

This table reports the distribution of bootstrapped estimates of the coefficient β from Eq. (3). For each pseudo-event date between January 1, 2017, and December 31, 2018, we construct a pseudo-sample spanning 180 days centered on the pseudo-event. Stocks are sorted based on pre–pseudo-event characteristics, and $Post_t$ is defined as an indicator equal to one for the post–pseudo-event period. We then estimate Eq. (3), interacting $Post_t$ with indicator variables capturing stock characteristics. The indicators considered are: $HighS_i$, equal to one for stocks in the top tercile of stock prices over the three months prior to the pseudo-event; $LowIV_i$, equal to one if the volume-weighted average implied volatility is in the bottom tercile; $LowO/S_{it}$, equal to one for stocks with a low ratio of option prices to stock prices; and $HighOmega_{it}$, equal to one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. Bootstrapped samples that overlap the actual event date are excluded. Panel A reports the coefficient estimates obtained using the pseudo-event corresponding to the actual event date. Panel B reports the empirical distribution of the corresponding bootstrapped coefficients for each interaction term, labeled by the indicator used. All specifications include stock and date fixed effects.

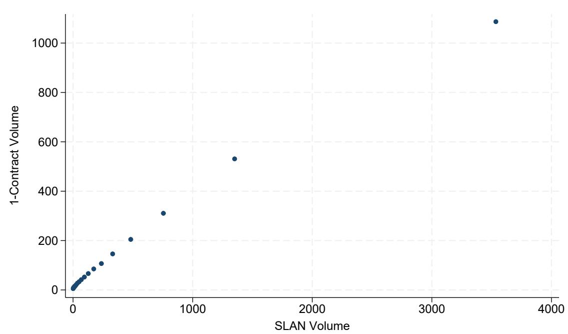
Panel A: Estimates				
	$HighS_i$	$LowIV_i$	$LowO/S_i$	$HighOmega_i$
Estimate	0.053	0.067	0.049	0.060

Panel B: Bootstrapped Distribution				
	$HighS_i$	$LowIV_i$	$LowO/S_i$	$HighOmega_i$
1	−0.046	−0.043	−0.049	−0.027
5	−0.043	−0.036	−0.042	−0.025
10	−0.028	−0.020	−0.028	−0.020
50	0.015	0.025	0.017	0.015
90	0.048	0.046	0.038	0.038
95	0.056	0.054	0.041	0.043
99	0.058	0.061	0.045	0.046
N	346	346	346	346

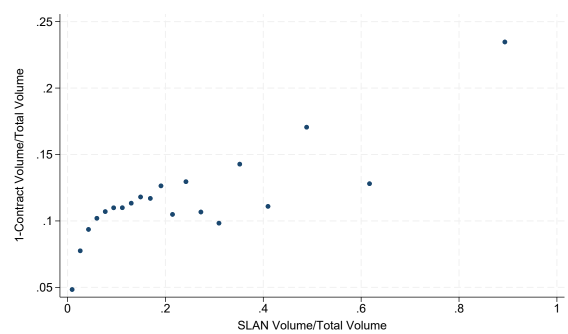
Figure A-1
Joint Distribution of Retail Option Trading Measures

This figure presents binscatter plots showing the joint distribution of measures of retail option trading. Each dot represents one twentieth of the sample, which is made up by stock-day level observations. Each stock-day observation aggregates the volume traded on all contracts that have that stock as underlying. Panels A and B (Panels C and D) plot the volume of one-contract option trades on the y-axis against the volume of retail trades on the x-axis, where retail trades are identified using the SLAN (all internalized) classification of Bryzgalova et al. (2023). Panels B and D report normalized measures obtained by scaling retail volumes by total option trading volume. Data on one-contract trades are obtained from CBOE. Data on SLAN-identified and internalized trades are from Bryzgalova et al. (2023), who use Options Price Reporting Authority (OPRA) data.

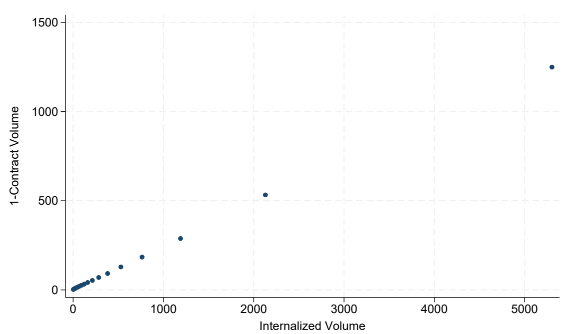
A: 1-contract vs. SLAN



B: 1-contract vs. SLAN (scaled)



C: 1-contract vs. All Internalized



D: 1-contract vs. All Internalized (scaled)

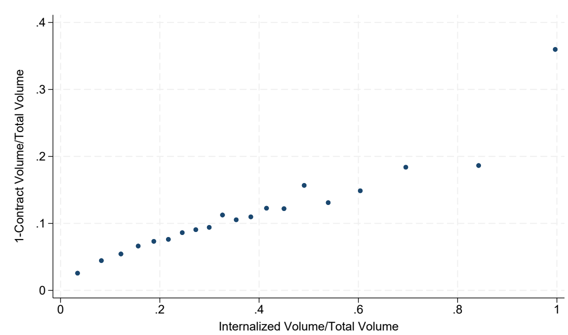
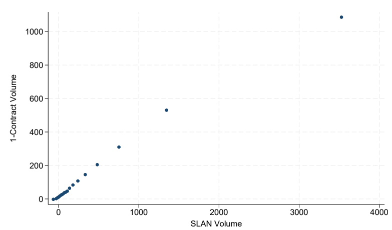


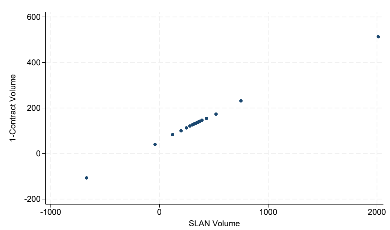
Figure A-2
Joint Distribution of Retail Option Trading Measures

This figure presents binscatter plots illustrating the joint distribution of alternative measures of retail option trading. Each dot represents one twentieth of the sample, aggregated at the stock–day level. For each stock–day, trading volume is aggregated across all option contracts written on the underlying stock. Panels A–D (Panels E–H) plot the volume of one-contract option trades on the y-axis against the volume of retail option trades on the x-axis, where retail trades are identified using the SLAN (all internalized) classification of Bryzgalova et al. (2023). Panels C, D, G, and H report normalized measures obtained by scaling retail trading volume by total option trading volume. Measures in Panels A, C, E, and G (Panels B, D, F, and H) are orthogonalized with respect to date fixed effects (firm fixed effects). Data on one-contract option trades are obtained from CBOE. Data on SLAN-identified and internalized trades are from Bryzgalova et al. (2023), who use Options Price Reporting Authority (OPRA) data.

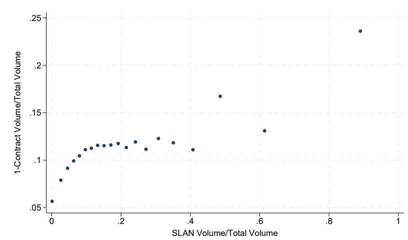
A: 1–SLAN (⊥ date)



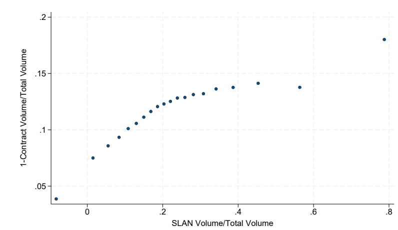
B: 1–SLAN (⊥ firm)



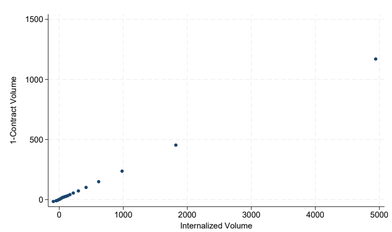
C: 1–SLAN (scaled, ⊥ date)



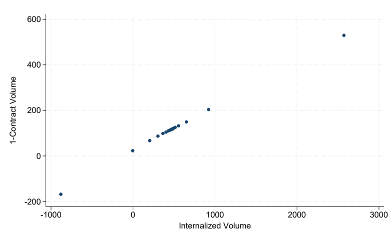
D: 1–SLAN (scaled, ⊥ firm)



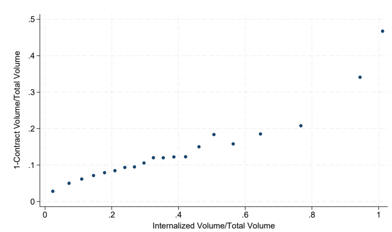
E: 1–All Int. (⊥ date)



F: 1–All Int. (⊥ firm)



G: 1–All Int. (scaled, ⊥ date)



H: 1–All Int. (scaled, ⊥ firm)

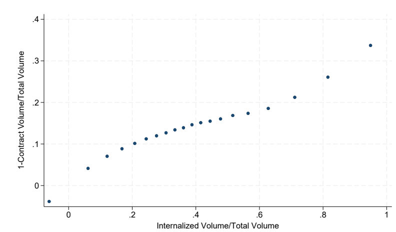
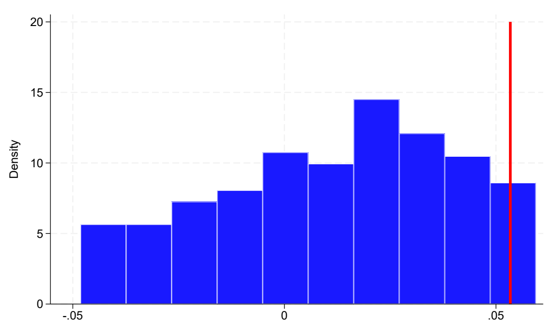


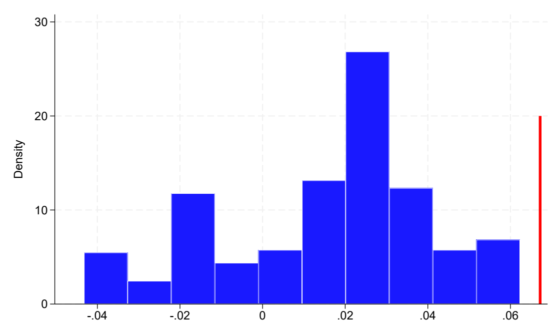
Figure A-3
Bootstrapped Distribution of Difference-in-Differences Estimates

This figures reports the distribution of bootstrapped estimates of the coefficient β from Eq. (3). For each pseudo-event date between January 1, 2017, and December 31, 2018, we construct a pseudo-sample spanning 180 days centered on the pseudo-event. Stocks are sorted based on pre–pseudo-event characteristics, and $Post_t$ is defined as an indicator equal to one for the post–pseudo-event period. We then estimate Eq. (3), interacting $Post_t$ with indicator variables capturing stock characteristics. The indicators considered are: $HighS_i$, equal to one for stocks in the top tercile of stock prices over the three months prior to the pseudo-event; $LowIV_i$, equal to one if the volume-weighted average implied volatility is in the bottom tercile; $LowO/S_{it}$, equal to one for stocks with a low ratio of option prices to stock prices; and $HighOmega_{it}$, equal to one for stocks in the top tercile of option leverage, defined as $\Delta S/O$. Bootstrapped samples that overlap the actual event date are excluded.Each panel plots the empirical distribution of the corresponding bootstrapped coefficients for each interaction term, labeled by the indicator used. All specifications include stock and date fixed effects.

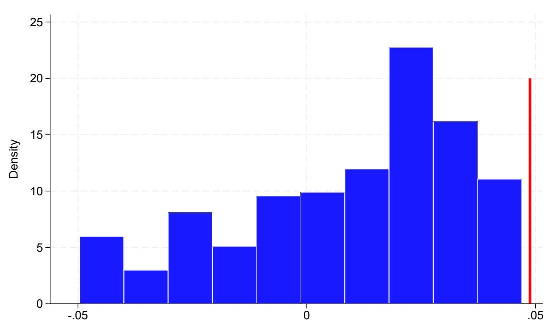
A: $HighS_i$



B: $LowIV_i$



C: $LowO/S_i$



D: $HighOmega_i$

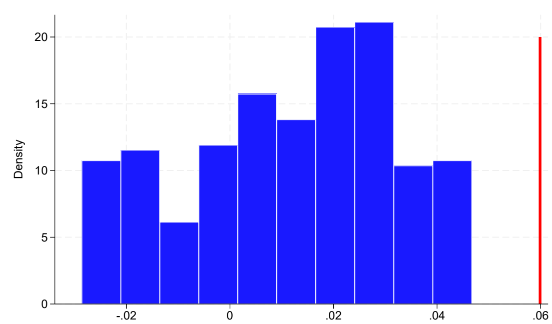


Figure A-4
Option Prices and Gamma

This figures shows the relation between an option's gamma and its price. We plot a binscatter showing the average option gamma for 50 equal-size buckets based on contract price. The sample consists of 10,612,980 daily observations for 1,327,066 option contracts from June 1, 2017, to May 31, 2018.